Topological Causal Data Analysis: A Framework for Integrating Persistent Homology with Causal Inference

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Abstract

Causal inference enables learning of intervention effects from data, yet modern scientific outcomes are increasingly complex, high-dimensional, and non-Euclidean, undermining the parametric and structural assumptions on which classical causal methods rely. Topological data analysis (TDA) studies the multi-scale shape of data through persistent homology and related functorial constructions, offering descriptors that are provably stable under perturbation but lacking native causal semantics. Despite a growing body of work at their interface—ranging from topologies on the space of structural causal models, to topology-aware treatment-effect estimators, to causal discovery via Hodge decomposition and topological orderings—no unified framework currently organizes these contributions. This paper proposes *Topological Causal Data Analysis* (TCDA) as a coherent research domain at the intersection of TDA and causal inference. We provide (i) a formal definition of a TCDA problem as a tuple linking a topological space of observations, a causal graph, a topological functor, and a causal query, together with foundational stability and identifiability propositions; (ii) a taxonomy organising existing and prospective work into four domains—topological causal representation, topological robustness in causal inference, causal interpretation of topological features, and methodological applications; (iii) a structured literature review and a suite of five proof-of-concept case studies that instantiate the framework on biomedical imaging, multivariate time series, confounding diagnostics, Mapper-based abstractions, and climate attribution; and (iv) a research agenda comprising eleven formal open questions, fourteen research avenues, and four computational challenges that span theory, computation, applications, and industrial implementation. The paper is designed to be self-contained for researchers from either TDA or causal inference, and aims to catalyse a decade of interdisciplinary work across statistics, machine learning, and the application domains where the shape of data and the structure of cause are intertwined.

1 Introduction

Topological data analysis (TDA) and causal inference (CI) are the two mathematical languages in which the modern data-analytic enterprise speaks most precisely about *shape* and about *consequence*. TDA, developed since the late 1990s, supplies a multi-scale, functorial, and provably stable grammar for the geometric structure of data [23, 29, 46, 47, 151]. Causal inference, developed in parallel since the 1970s, supplies a complete calculus for the interventional and counterfactual content of probabilistic models [10, 73, 113, 114, 118, 124]. Both fields have reached methodological maturity; both have generated their own journals, their own statistical scaffolding, and their own implementation ecosystems; and both have been applied to many of

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the same empirical objects—networks, dynamical systems, biomedical images, spatiotemporal fields—without explicitly meeting.

The thesis of this paper is that the absence of an explicit meeting between the two is no longer tenable. Modern scientific data are increasingly complex, high-dimensional, and non-Euclidean, in ways that undermine the parametric and structural assumptions on which classical causal estimators depend; and topological data analysis, having absorbed a decade of statistical inference results, is now positioned to deliver not merely descriptive summaries but *causal* estimators with calibrated uncertainty. A small but growing body of work—the four seed papers [48, 72, 82, 89] that we will treat as the anchors of the field, together with the sheaf-theoretic causal-inference programme [96–98] and the topological-ordering family of causal-discovery algorithms [76, 123, 144, 153, 154]—has begun to formalise the connection. What is missing is the framework that organises these contributions into a single research domain.

We propose such a framework. The remainder of this introduction positions the contribution: Section 1.1 traces the parallel evolution of TDA and CI; Section 1.2 argues for their integration; Section 1.3 states the limitations of each tradition in isolation; and Section 1.4 enumerates the four contributions that we will make in the body of the paper.

1.1 The parallel evolution of two fields

Topological data analysis crystallised as a discipline with the introduction of *persistent homology*: the assignment, to a filtered simplicial complex, of a multi-set of birth–death pairs that record the appearance and disappearance of topological features at each scale [45, 47, 156]. The stability theorem of Cohen-Steiner et al. [38], refined by Chazal et al. [34], established that persistent homology is 1-Lipschitz with respect to the bottleneck distance under L^∞ perturbations of the filter, turning a previously qualitative summary into a calibrated descriptor. The decade that followed delivered the statistical inference programme—confidence sets, bootstrap, rates of convergence, and functional summaries (landscapes, silhouettes, images)—synthesised in Lecci [89] and consolidated by Adams et al. [1], Bubenik [20], Chazal et al. [33]. Carlsson [23], the Singh et al. [128] Mapper algorithm, and the categorical formalism of Curry [40], Spivak [136] situated TDA within a broader functorial-data-analysis programme that emphasised composability and structure.

Causal inference, in parallel, developed in two complementary traditions. Rubin [124] and the potential-outcomes school [73, 125] grounded the field in counterfactual contrasts and identification conditions (consistency, ignorability, positivity), and produced the doubly-robust efficient estimation machinery [35, 81, 120] that today underwrites empirical work in biomedicine, econometrics, and policy science. Pearl [113, 114], Pearl and Mackenzie [116] introduced the structural-causal- model formalism and do-calculus, which axiomatised intervention as a syntactic operation on directed acyclic graphs and enabled *causal discovery*: the recovery of G itself from data via constraint-based [135], score-based [100], and functional [117, 127] methods. Bareinboim et al. [10], Ibeling and Icard [72] crystallised the causal hierarchy (observational, interventional, counterfactual) into a single expressive ladder.

Figure 3 in Section 4 traces the two trajectories side by side. Until ∼2020 there is essentially no crossing: a researcher could attend the major venues of either field for a decade without hearing the other's vocabulary. The crossings that began to appear thereafter—a topology on the space of SCMs [72]; topology-aware treatment-effect estimators [82]; simulation-based topological inference for multivariate time series [48]; sheaf-theoretic causal calculi [96–98]; and topological-ordering algorithms for causal discovery [76, 123, 144, 153, 154]—are the methodological seeds of the field this paper proposes to name.

1.2 Why topology matters for causality

Three properties of topology recommend it as a natural partner for causal inference, and each is reflected in one of the four domains of our framework (Figure 1).

Non-Euclidean data. The classical average treatment effect, $\mathrm{ATE} = \mathbb{E}[Y^1 - Y^0]$, implicitly assumes that the difference $Y^1 - Y^0$ is well defined and scalar-valued. Modern outcomes routinely fail both assumptions: a CT scan, a molecular conformation, a brain connectivity graph, and a gridded climate field are all non-Euclidean objects on which "$Y^1 - Y^0$" has no obvious meaning. Topological summaries of such objects—persistence diagrams, Reeb graphs, Mapper complexes—live in calibrated metric spaces (the Wasserstein space of diagrams, the interleaving space of Reeb graphs) on which causal contrasts are well defined [82, 88]. Sub-domain A3 of Section 3 (causal embeddings in persistence diagrams) makes this construction precise.

Robustness through stability. The stability theorem (Equation (4)) is the operational core of the field: small perturbations in the data produce only small perturbations in the persistent invariants, with a constant of one. This is precisely the kind of bound that classical causal estimators typically lack and that practical deployment increasingly demands. Proposition 3.1 shows that any Lipschitz causal functional of a persistent invariant inherits the quantitative robustness of PH_k, giving a TCDA estimator a tractable sensitivity guarantee with respect to Hausdorff-scale perturbations of the data support. Sub-domain B1 of Section 3 formalises this transfer.

Hidden structure and confounding. Unobserved confounders move observed variables along unobserved orbits and so deposit a topological signature in the joint support that no graphical or moment-based diagnostic detects. The worked example of Section 5.3—a hidden circular confounder generating a persistent 1-cycle in the residual support of an apparently null treatment effect—illustrates the principle, and Proposition 5.1 states it formally. Sub-domain B2 of Section 3 positions topological diagnostics as a complement to the meagre-set theorem of Ibeling and Icard [72]: where the theorem tells us that assumption-free identification is impossible on almost all SCMs, topological diagnostics tell us *which* SCMs have residual geometric structure consistent with hidden confounding.

These three properties are sufficient to motivate the framework, but the case for TCDA also rests on a fourth, less elementary observation: much of modern causal discovery is *already* doing topological data analysis, often without naming it as such. Jiang et al. [76], Rolland et al. [123], Tran et al. [144], Wu and Xie [153], Xu et al. [154] all extract a topological order from the data and then refine edges within that order; the order itself is an invariant of the joint distribution under the appropriate functor, and Proposition 3.2 formalises the corresponding identifiability question. Naming what is already implicit is a small contribution, but it is one that the field requires before it can begin to make explicit progress on the deeper questions.

1.3 Limitations of current approaches

The case against the status quo is symmetric. Classical causal inference, applied to non-Euclidean outcomes, must reduce them to scalar surrogates and so discards exactly the structure of interest; classical TDA, applied to data with a known causal-graph context, exposes geometric features but is silent on the question of which geometric feature carries an intervention's effect. Neither field, on its own, is equipped to handle a treatment whose outcome is a tumour shape, a folded protein, a brain network, or a precipitation cluster.

A further limit is structural rather than technical. Ibeling and Icard [72]'s topological causal hierarchy theorem shows that the set of SCMs on which assumption-free identification of $P(y \mid \mathrm{do}(x))$ is possible is *topologically meagre*—in the Baire-category sense, exceptional. This is

the topological counterpart of measure-theoretic "no-free-lunch" results [15, 55, 57, 110], and it places a precise upper limit on what any purely observational causal procedure can hope to achieve. The theorem also identifies the structure that must be supplied to do better, namely a topology on the model space, and is one of the clearest demonstrations that the language of CI requires topological extension on its own terms. The complementary critique runs from CI back to TDA: a persistence diagram is the same whether the variables it summarises stand in a causal relationship or are mere coincident features, so without an intervention semantics overlay, a TDA descriptor cannot license a causal claim. Section 4.4 formalises five specific gaps that follow from this asymmetric maturity, and Section 6 translates each into a concrete open problem.

1.4 Our contributions

The paper makes four contributions.

(1) **A formal framework.** We propose *Topological Causal Data Analysis* (TCDA) as a coherent research domain at the intersection of TDA and CI, give a formal definition of a TCDA problem as a tuple $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ (Definition 3.1), state two foundational propositions (Propositions 3.1 and 3.2) that establish well-posedness, and defend a taxonomy of four top-level domains and twelve sub-domains (Figure 1). This is the central intellectual contribution and is the subject of Section 3.

(2) **A structured literature review.** Section 4 synthesises the existing connections between TDA and CI along the taxonomy of Section 3, organised across three eras (Sections 4.1 to 4.3) and closing with five concrete gaps (Section 4.4). The review establishes that every node of the taxonomy is already anchored in published work, and that the missing piece is the unifying framework we provide.

(3) **A suite of proof-of-concept case studies.** Section 5 instantiates the TCDA tuple in five concrete settings—biomedical imaging and molecular structure (Section 5.1), multivariate brain time series (Section 5.2), confounding diagnostics (Section 5.3), Mapper-based causal abstraction with calibrated uncertainty (Section 5.4), and spatiotemporal climate attribution (Section 5.5)—together with a worked confounding-signature proposition (Proposition 5.1) and a synthesis table (Table 1) mapping each case study back to the taxonomy.

(4) **A research agenda.** Section 6 formalises eleven Open Questions (Open Question 6.1–Open Question 6.11), fourteen Research Avenues (Research Avenue 6.1–Research Avenue 6.14), and four Computational Challenges (Challenge 6.1–Challenge 6.4), organised into theoretical, methodological, computational, domain-specific, and industry-facing branches and summarised in the mind-map of Figure 4. Each item is calibrated to be specific enough to underwrite an individual research project or grant proposal.

What this paper is not. TCDA is a proposal, not a theorem. We do not claim that every component of the framework is mature, that every open question is solvable, or that the taxonomy is final. Several of the propositions in Sections 3 and 5 are stated as proof sketches rather than full theorems, and several of the open questions in Section 6 are likely to be reformulated as the field matures. The contribution of the paper is to articulate the framework, locate the existing literature within it, and identify the questions whose answers will determine whether TCDA becomes a durable research domain or remains a useful but provisional re-organisation of work that already exists.

Organisation. Section 2 collects the minimal TDA and CI background needed to read the rest of the paper. Section 3 develops the TCDA framework and proves its foundational propositions. Section 4 surveys the existing state of the field. Section 5 presents five proof-of-concept case

studies. Section 6 states the research agenda. Section 7 closes with a call for interdisciplinary collaboration.

2 Preliminaries

This section assembles the minimal background needed to read the rest of the paper. The exposition deliberately addresses two audiences in parallel: readers fluent in topological data analysis (TDA) who have only passing familiarity with formal causal inference (CI), and readers fluent in CI who have only passing familiarity with persistent homology. We therefore strive for completeness over compression and fix the notation that Sections 3 to 6 will reuse.

2.1 Topological data analysis

TDA studies the multi-scale geometric and topological structure of data using tools from algebraic topology [23, 29, 46, 151]. The unifying idea is *functoriality*: a data set is mapped, in a stable and computable way, to an algebraic object whose invariants summarise its shape.

2.1.1 Simplicial complexes and filtrations

Let V be a finite vertex set. An abstract simplicial complex on V is a collection $K \subseteq 2^V$ closed under taking subsets: if $\sigma \in K$ and $\tau \subseteq \sigma$, then $\tau \in K$. A simplex $\sigma \in K$ with $|\sigma| = k+1$ is called a k-simplex; the dimension of K is the maximum dimension of its simplices. A 1-dimensional simplicial complex is a graph; higher-dimensional simplicial complexes admit triangles, tetrahedra, and so on.

Given a finite point cloud $X \subset (\mathcal{X}, d)$ in a metric space and a scale parameter $r \geq 0$, the *Vietoris–Rips complex* is

$$\mathrm{Rips}_r(X) = \{\sigma \subseteq X : d(u, u') \leq 2r \text{ for all } u, u' \in \sigma\}, \tag{1}$$

and the *Čech complex* $\check{\mathrm{C}}_r(X)$ collects subsets whose closed r-balls have non-empty common intersection [67]. For $X \subset \mathbb{R}^m$ the *α-complex* $\mathrm{Alpha}_r(X)$ is built from intersections of Voronoi cells with r-balls and yields a homotopy-equivalent but smaller complex than $\check{\mathrm{C}}_r$. For image-like grid data, *cubical complexes* play the analogous role, with elementary cubes replacing simplices.

A *filtration* of K is a nested family $\mathcal{F} = \{K(t)\}_{t \in \mathbb{R}}$ with $K(s) \subseteq K(t)$ whenever $s \leq t$. Increasing r in eq. (1) produces a Vietoris–Rips filtration; sublevel sets $K(t) = f^{-1}((-\infty, t])$ of a monotone function $f : K \to \mathbb{R}$ produce a sublevel filtration. Filtrations expose the data's structure at multiple resolutions simultaneously.

2.1.2 Homology and persistent homology

Singular or simplicial homology assigns to each topological space (or simplicial complex) X a graded sequence of abelian groups $H_k(X)$, where H_0 counts connected components, H_1 counts independent 1-dimensional loops, H_2 counts enclosed voids, and so on; the rank of $H_k(X)$ is the kth *Betti number* $\beta_k(X)$ [66, 105]. Homology is a functor: a continuous map $f : X \to Y$ induces a group homomorphism $H_k(f) : H_k(X) \to H_k(Y)$.

Applied to a filtration $\mathcal{F} = \{K(t)\}$, the inclusion $K(s) \hookrightarrow K(t)$ induces, for every k, a linear map $H_k(K(s)) \to H_k(K(t))$. The collection $\mathrm{PH}_k(\mathcal{F}) = \{H_k(K(t))\}_{t \in \mathbb{R}}$ together with these maps is the *k-th persistent homology module* of $\mathcal{F}$ [45, 47, 156]. Over a field, any tame persistent homology module decomposes uniquely (up to isomorphism) into a direct sum of interval modules, and the corresponding intervals $[b, d) \subset \mathbb{R}$ form the *persistence barcode*; equivalently one may record their endpoints as a multiset of points

$$\mathrm{Dgm}_k(\mathcal{F}) \subset \mathbb{R}^2_+ := \{(b, d) \in (\mathbb{R} \cup \{\infty\})^2 : b < d\}, \tag{2}$$

the *persistence diagram* in homology degree k [46, 60]. Following standard convention, Dgm_k includes the diagonal $\Delta = \{(t,t) : t \in \mathbb{R}\}$ with infinite multiplicity, so matchings between diagrams may send unmatched points to Δ.

2.1.3 Distances on diagrams and the stability theorem

The two principal metrics on the space of persistence diagrams are the *bottleneck distance* d_{B} and the p-Wasserstein distance W_p:

$$d_{\mathrm{B}}(D, D') = \inf_{\gamma} \sup_{x \in D} \|x - \gamma(x)\|_\infty, \qquad W_p(D, D') = \Big(\inf_{\gamma} \sum_{x \in D} \|x - \gamma(x)\|_\infty^p\Big)^{1/p}, \tag{3}$$

where γ ranges over bijections $D \cup \Delta \to D' \cup \Delta$. The cornerstone of statistical TDA is the *stability theorem*: if f, g are tame functions on a triangulable space, then

$$d_{\mathrm{B}}(\mathrm{Dgm}_k(f), \mathrm{Dgm}_k(g)) \le \|f - g\|_\infty, \tag{4}$$

and analogous bounds hold in Wasserstein distance [34, 38]. Stability is the property that makes persistent homology a viable summary for noisy data and is what will allow us, in Section 3, to lift TDA-level perturbation guarantees to causal-inference-level robustness statements.

2.1.4 Functional summaries: landscapes, silhouettes, Euler curves

For statistical and machine-learning use, it is convenient to embed Dgm_k in a Hilbert or Banach space. Three standard embeddings will appear repeatedly. For $p = (a, b) \in \mathrm{Dgm}_k$, define the tent function $\Lambda_p(t) = \max\{0, \min(t - a, b - t)\}$. The *persistence landscape* $\{\lambda_j(t; D)\}_{j \ge 1}$ records the j-th largest value of $\{\Lambda_p(t)\}_{p \in D}$ [20]. The *power-weighted silhouette* of order $r > 0$ is

$$\varphi(t; D, r) = \frac{\sum_{p=(a,b) \in D} (b - a)^r \, \Lambda_p(t)}{\sum_{p=(a,b) \in D} (b - a)^r}, \qquad t \in T \subset \mathbb{R}, \tag{5}$$

with T a fixed compact interval [33, 82]. The *persistence image* [1] provides a fixed-dimensional vector embedding. Finally, the *Euler characteristic curve* records $\chi(K(t)) = \sum_k (-1)^k \beta_k(K(t))$ along the filtration; though it discards information, it is cheap to compute and frequently appears in applications [131, 132].

Lemma 2.1 (Silhouette Lipschitz stability; informal restatement of 82)**.** *Under a uniform bound on tent heights, the power-weighted silhouette satisfies* $\|\varphi(\cdot; D) - \varphi(\cdot; D')\|_\infty \le c_r \, W_1(D, D')$ *for an explicit constant* $c_r > 0$ *depending only on* r *and an upper bound on persistence lifetimes.*

Lemma 2.1 is the bridge by which Wasserstein perturbations of diagrams translate into sup-norm perturbations of silhouettes; Section 3 uses it to derive causal-stability statements.

2.1.5 Mapper and Reeb graphs

The Mapper algorithm [43, 95, 128] produces a combinatorial summary of a high-dimensional point cloud X. Given a *filter* (or lens) function $f : X \to \mathbb{R}^d$, a cover $\{U_\alpha\}$ of $f(X)$, and a clustering procedure, Mapper assigns one node per cluster of $f^{-1}(U_\alpha)$ and connects nodes whose clusters intersect. The resulting simplicial complex is a discrete proxy for the Reeb graph of f, which collapses connected components of level sets to points [14, 16, 102]. Reeb graphs and Mapper are central in the topological abstraction of causal mechanisms (Section 3, Domain A2).

2.2 Causal inference

Causal inference seeks to quantify what would happen to an outcome under hypothetical interventions on a system, going beyond the associations captured by joint distributions [73, 114, 118, 135]. Two complementary frameworks dominate the methodological landscape.

2.2.1 Structural causal models

A *structural causal model* (SCM) is a tuple $\mathcal{M} = \langle U, V, \{f_V\}_{V \in V}, P \rangle$, where V is a set of *endogenous* variables, U a set of *exogenous* variables, $f_V : \chi_{\mathrm{Pa}(V)} \times \chi_{U(V)} \to \chi_V$ is a measurable structural function determining each endogenous variable from its parents and exogenous noise, and P is a Borel probability measure on χ_U [17, 72, 114]. Each SCM induces a unique observational distribution $p^{\mathcal{M}} \in \mathbb{P}(\chi_V)$ under recursiveness assumptions; under an *intervention* $W := w$ on a subset $W \subseteq V$, the intervened SCM $\mathcal{M}_{W:=w}$ holds $W = w$ fixed and otherwise leaves mechanisms intact, yielding an *interventional* distribution $p^{\mathcal{M}_{W:=w}}$ [114].

Pearl's do-calculus [90, 113, 143] provides a complete set of rewrite rules for the intervention operator $\mathrm{do}(\cdot)$ on a directed acyclic graph (DAG); it permits the algorithmic identification of interventional from observational distributions whenever such identification is possible. The *causal hierarchy*—observational, interventional, counterfactual— organises causal queries by the strength of assumption needed to identify them [10, 72].

A *causal graph* $G = (V, E)$ on the variable set V encodes the direct-influence relation: $X \to Y \in E$ means $X \in \mathrm{Pa}(Y)$. We write G for a generic causal graph and reserve $\mathcal{M}$ for SCMs whose graph structure is G. The space of all SCMs compatible with G is denoted $\mathfrak{M}_G$; without graph restriction we write $\mathfrak{M}$. The topology on $\mathfrak{M}$ used in Ibeling and Icard [72] is the weak topology induced by the convergence of the joint distribution over all interventions, and is one of the central objects in Section 3, Domain B.

2.2.2 Potential outcomes and causal estimands

The *potential outcomes* (or Neyman–Rubin) framework [73, 124, 125] parameterises causal effects directly in terms of counterfactual outcomes. For a binary treatment $A \in \{0, 1\}$ and observed outcome Y, one posits counterfactual outcomes Y^0, Y^1 representing the response under each treatment. The *average treatment effect* is

$$\mathrm{ATE} = \mathbb{E}[Y^1 - Y^0], \tag{6}$$

and conditioning on covariates X yields the *conditional average treatment effect* $\mathrm{CATE}(x) = \mathbb{E}[Y^1 - Y^0 \mid X = x]$. Identification of eq. (6) from observational data typically rests on three assumptions:

(C1) *Consistency*: $Y = Y^A$ almost surely.

(C2) *No unmeasured confounding* (a.k.a. ignorability): $\{Y^0, Y^1\} \perp\!\!\!\perp A \mid X$.

(C3) *Positivity (overlap)*: $0 < \mathbb{P}(A = 1 \mid X) < 1$ almost surely.

Under (C1)–(C3), the ATE is identified by $\mathrm{ATE} = \mathbb{E}[\mu_1(X) - \mu_0(X)]$ with $\mu_a(x) = \mathbb{E}[Y \mid X = x, A = a]$, and is estimable by inverse-probability weighting, outcome regression, or *augmented inverse-probability weighting* (AIPW) [35, 81]. AIPW is *doubly robust*: consistency holds if either the outcome or propensity model is consistent, and rate-double-robust efficiency holds under product-rate nuisance conditions [80, 106].

When the outcome is functional or non-Euclidean, the ATE in eq. (6) is no longer well defined, and one must replace the difference of means by a metric- or geometry-respecting estimand. Two such generalisations recur in this paper:

- the *functional average treatment effect*, defined pointwise in a function space and estimated via doubly-robust AIPW on the function-on-scalar regression [44, 93, 140];
- the *geodesic average treatment effect*, defined in a geodesic metric space and estimated via Fréchet means [88].

Functional ATEs on *topological* summaries (such as silhouettes) are exactly what Kim and Lee [82] formalise, and we will see in Section 3 that this construction is one of three founding pillars of the TCDA framework.

2.2.3 Causal discovery

Beyond effect estimation, *causal discovery* attempts to learn the graph G itself from data [117, 127, 135, 150]. Methods fall into three broad families:

- *Constraint-based*: query conditional independencies and prune candidate DAGs accordingly (PC, FCI).
- *Score-based*: define a regularised likelihood over DAGs and search the space (GES, NOTEARS).
- *Functional causal models*: exploit functional asymmetries such as additive non-Gaussian noise to identify the causal direction (LiNGAM, ANM).

A growing body of work uses *topological orderings* of variables to shrink the search space [11, 144, 153, 154]; this line is one of the most direct existing bridges between TDA and CI and is taken up in Section 3, Domain A1.

2.3 Existing mathematical bridges

The interface between TDA and CI is not a void: several mathematical threads already connect the two communities, although they have not yet been unified into a single domain. We review three.

2.3.1 Topologies on causal-model spaces

Following Dembo and Peres [42] and Genin and Kelly [57], open sets of the weak topology on probability distributions correspond to statistically *verifiable* hypotheses. Ibeling and Icard [72] extend this correspondence to the space of SCMs by constructing a sequence of topologies, one for each level of the causal hierarchy, and proving that the collapse set of the hierarchy is topologically meager. Earlier parametric results on faithfulness violations [100, 146] can be re-read in this light as measure-theoretic predecessors to the topological statement. We use the phrase "topology of causal models" for this line of work and adopt it as the foundation of Section 3, Domain B.

2.3.2 Sheaf-theoretic and categorical approaches

Sheaves, originally developed for cohomology and complex geometry [18], have re-emerged as a language for distributed data, signal processing, and now causal inference [40, 122, 136]. Mahadevan [96, 97, 98] develop a topos-theoretic formulation of causal inference in which do-calculus is generalised via subobject classifiers, the space of causal DAGs is endowed with an Alexandroff topology, and certain identification results take the form of sheaf-cohomology statements. Independently, the categorical view of persistent homology—homology is a functor, and filtrations are functors from $(\mathbb{R}, \leq)$ to the category of topological spaces—suggests that any *functorial* causal inference procedure should commute, up to a controllable error, with the topological functor used to summarise data. This is the formal content of Domain B3 ("functorial causal inference") in Section 3.

2.3.3 Optimal transport and Wasserstein space

Optimal transport equips spaces of probability measures with rich geometric structure [119, 149]. Two threads concern this paper. First, persistence diagrams are themselves compared via W_p-distances; many statistical guarantees for persistent homology are expressed in these terms [33, 101, 145]. Second, the Wasserstein distance offers a natural metric on counterfactual distributions and so on causal effect estimators, with stability results connecting changes in observed distributions to changes in identified causal quantities. The combination of these two

threads underwrites the silhouette stability bound of Kim and Lee [82], which we will use in Section 3.

2.4 Notation summary

We collect the notation that recurs throughout the paper.

Symbol	Meaning
$(\mathcal{M}, d)$	Topological / metric space of observations (a Polish space, in the broadest setting).
$\mathcal{F} = \{K(t)\}_{t\in\mathbb{R}}$	A filtration of simplicial / cubical complexes built from data.
$\mathrm{PH}_k(\mathcal{F})$	k-th persistent homology module of the filtration.
$\mathrm{Dgm}_k(\mathcal{F})$	k-th persistence diagram (multiset of birth–death pairs).
$d_{\mathrm{B}}, W_\infty, W_p$	Bottleneck and p-Wasserstein distances on diagrams.
$\varphi(t; D, r)$	Power-weighted silhouette function of diagram D at exponent r.
$\mathrm{Map}_f, \mathrm{Reeb}_f$	Mapper and Reeb-graph summaries with filter f.
$G = (V, E)$	Causal graph (directed acyclic, unless stated).
$\mathcal{M} \in \mathfrak{M}_G$	SCM with graph G; $\mathfrak{M}$ if the graph is unspecified.
$p^{\mathcal{M}}, p^{\mathcal{M}_{W:=w}}$	Observational and interventional distributions of $\mathcal{M}$.
Y^a	Potential outcome under treatment $A = a$.
ATE, CATE, TATE, GATE	Average / conditional / topological / geodesic ATEs.
$\mathcal{T}$	A topological functor (persistent homology, Mapper, etc.) acting on data.
$\mathcal{C}$	A causal query (e.g., ATE, do-expression, identifiability question).
TCDA	Topological Causal Data Analysis (the proposed domain).

This notation is used uniformly in Section 3 onwards. The quadruple $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ in particular will be the carrier of the TCDA formalism.

Equipped with these tools, we now turn to the main intellectual contribution of the paper.

3 The TCDA Framework

Building on the preliminaries of Section 2, we now propose *Topological Causal Data Analysis* (TCDA) as a coherent research domain at the intersection of TDA and causal inference. Our exposition proceeds in five steps. Section 3.1 gives a formal definition of a TCDA problem as a tuple linking a topological space of observations to a causal graph through a functorial topological summary, together with two foundational propositions establishing well-posedness. Sections 3.2 to 3.5 then defend a four-domain taxonomy: representation (A), robustness (B), interpretation (C), and methodology (D), each with three sub-domains. The taxonomy is summarised in Figure 1 and is the paper's primary intellectual contribution. Every sub-domain is anchored to at least one published work, ensuring that the framework is grounded rather than aspirational.

3.1 Formal definition of a TCDA problem

The unifying intuition is that, in a TCDA problem, causal inference is performed not on raw observations but on *topological summaries* of those observations, where the summary is computed by a functor that satisfies the stability and interpretability properties recalled in Section 2.1. The causal question of interest is then either (i) a query about the underlying causal structure that generated the observations, or (ii) a contrast between topological summaries induced by different interventions, or (iii) a robustness/identifiability statement about classical causal estimands whose validity is controlled by topological invariants.

Definition 3.1 (TCDA problem)**.** *A* topological causal data analysis problem *is a tuple*

$$\text{TCDA} \;=\; (\mathcal{M},\, G,\, \mathcal{T},\, \mathcal{C}),$$

whose components are:

(i) $\mathcal{M}$ *is a (Polish) topological space of observations equipped with a metric or filtration parameter* $\rho : \mathcal{M} \times \mathcal{M} \to \mathbb{R}_{\geq 0}$*, on which probability measures and their interventional counterparts are well-defined;*

(ii) $G = (V, E)$ *is a causal structure—typically a DAG or ADMG—whose vertices represent (possibly latent) random variables* $X = (X_v)_{v \in V}$ *taking values in* $\mathcal{M}$ *and whose edges encode direct causal relationships in the SCM sense of Section 2.2;*

(iii) $\mathcal{T} : \mathbf{Filt}(\mathcal{M}) \to \mathbf{Pers}$ *is a* topological functor *from a category of filtered objects on* $\mathcal{M}$ *to a category of persistence-like invariants. Canonical instances are* $\mathcal{T} = \mathrm{PH}_k$ *for* $k \in \{0, 1, 2\}$*,* $\mathcal{T} = \mathrm{Reeb}_f$*,* $\mathcal{T} = \mathrm{Map}_f$*, and their functional summaries* $\mathcal{T} = \varphi(\cdot; r)$ *(silhouettes) or* $\mathcal{T} = \Lambda$ *(landscapes). We require* $\mathcal{T}$ *to be stable with respect to some interleaving-type metric on* $\mathbf{Filt}(\mathcal{M})$*;*

(iv) $\mathcal{C}$ *is a* causal query*—a functional of the interventional or counterfactual distribution induced by* G *on* $\mathcal{M}$*. We allow* $\mathcal{C}$ *to act directly on raw outcomes (*$\mathcal{C} = \mathrm{ATE}, \mathrm{CATE}, \mathrm{TATE}, P(y \mid \mathrm{do}(x)), \ldots$*) or on topological summaries (*$\mathcal{C} = W_p(\mathrm{Dgm}_k(P^x), \mathrm{Dgm}_k(P^{x'}))$*, for example, as in Kim and Lee [82]).*

A TCDA solution *is an estimator* $\widehat{\mathcal{C}}_n$ *together with calibrated uncertainty (confidence sets, hypothesis tests, posterior credible sets) for the population quantity* $\mathcal{C}$*, computed from a sample* $\{X_i\}_{i=1}^n \subset \mathcal{M}$ *using* $\mathcal{T}$ *as a (possibly sufficient) statistic.*

Definition 3.1 is intentionally broad. The four components $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ play orthogonal roles: $\mathcal{M}$ specifies *what the data is*, G specifies *which causal relationships are admitted*, $\mathcal{T}$ specifies *how shape is summarised*, and $\mathcal{C}$ specifies *the scientific question*. Existing work occupies distinct corners of this space:

- Ibeling and Icard [72] take $\mathcal{T}$ to be the identity and endow $\mathcal{M} = \mathfrak{M}$ (a space of SCMs) with a topology that controls the Pearl–Bareinboim causal hierarchy.
- Kim and Lee [82] fix $\mathcal{C}$ to be a Wasserstein contrast between persistence diagrams of potential outcomes and develop AIPW estimators with $\sqrt{n}$-rates and weak-convergence guarantees.
- El-Yaagoubi et al. [48] fix $\mathcal{M}$ to be a space of multivariate time-series, $\mathcal{T} = \mathrm{PH}_1$ of a windowed dependence network, and $\mathcal{C}$ to be a hypothesis test about the number of cycles.
- Lecci [89] develops the calibrated statistical inference for $\mathcal{T}$ alone that any TCDA estimator must inherit.

Thus the four seed works of the field each fix three components of the TCDA tuple and study the remaining one. The TCDA programme is precisely the systematic study of the joint behaviour of all four.

Requirements on $\mathcal{T}$. For Definition 3.1 to be useful in a causal context, the functor $\mathcal{T}$ must satisfy three operating conditions:

(F1) *Stability.* Small perturbations of the input filtration should produce small perturbations of $\mathcal{T}(\cdot)$ in an interleaving-type metric. Persistent homology satisfies this with constant 1 for the bottleneck distance (Equation (4)). Mapper and Reeb functors admit weaker but useful stability properties under appropriate discretisation [26, 104].

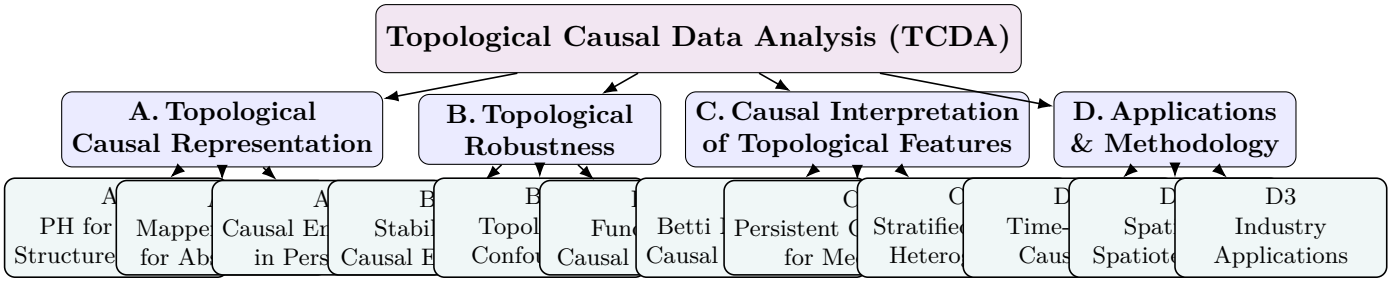

Figure 1: The proposed TCDA taxonomy. Four orthogonal domains, each with three sub-domains, organise the field. Domain A concerns *representation*: how topology summarises the structure of causal data. Domain B concerns *robustness*: how topological stability controls the sensitivity of causal estimates to perturbations. Domain C concerns *interpretation*: which causal stories are encoded by topological features. Domain D concerns *methodology and applications* in concrete data modalities. The taxonomy is intended to be exhaustive (every existing TCDA contribution we are aware of fits in at least one node) but not mutually exclusive (many works contribute to several sub-domains simultaneously).

(F2) *Interpretability.* The features of $\mathcal{T}(\cdot)$ should admit a substantive interpretation in the underlying scientific domain (e.g., 1-cycles in PH_1 of a brain functional network correspond to re-entrant cortical loops; 48, 129).

(F3) *Computability.* The functor must be computable in time polynomial in the sample size for the kind of data the application demands. Off-the-shelf libraries `Ripser`, `GUDHI`, `giotto-tda` provide $O(n^3)$ persistent-homology computation up to dimension 2 in practice; sparse and approximate variants extend this to $n \gtrsim 10^5$ [13, 109, 139].

We refer to functors satisfying (F1)–(F3) as *admissible* and restrict attention to admissible $\mathcal{T}$ in all that follows.

We now state two propositions that establish the well-posedness of the TCDA tuple. The first transfers the celebrated stability of persistent homology to any Lipschitz causal query; the second pins down the informational condition under which a causal graph is identifiable from a topological functor alone. Both are stated for the canonical case $\mathcal{T} = \mathrm{PH}_k$ but extend immediately to silhouettes, landscapes, and other Lipschitz functional summaries.

Proposition 3.1 (Stability transfer)**.** *Let* TCDA $= (\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ *be a TCDA problem with* $\mathcal{T} = \mathrm{PH}_k$ *applied to the Vietoris–Rips filtration of a compact subset of* $\mathcal{M}$*, and let*

$$\mathcal{C}[\mathcal{T}(P), \mathcal{T}(Q)] \;=\; \phi(\mathrm{Dgm}_k(P),\, \mathrm{Dgm}_k(Q))$$

for some functional ϕ *that is* L*-Lipschitz with respect to the* p*-Wasserstein distance on persistence diagrams. Then for any pair of distributions* $P, P' \in \mathcal{P}(\mathcal{M})$ *with bounded support and any fixed* Q*,*

$$|\mathcal{C}[\mathcal{T}(P), \mathcal{T}(Q)] - \mathcal{C}[\mathcal{T}(P'), \mathcal{T}(Q)]| \;\le\; L \cdot W_p(\mathrm{Dgm}_k(P),\, \mathrm{Dgm}_k(P')) \;\le\; L \cdot d_H(\mathrm{supp}\, P,\, \mathrm{supp}\, P'),$$

where d_H *denotes the Hausdorff distance between the supports of* P *and* P'*.*

Sketch. The first inequality is the Lipschitz assumption on ϕ; the second is the stability theorem for persistent homology (Equation (4); 34, 38). The constant is 1 in the canonical bottleneck case and tracks the p-Wasserstein constant otherwise. □

Significance. Proposition 3.1 says that any causal query expressible as a Lipschitz functional of persistence diagrams inherits the quantitative robustness of PH_k: a small Hausdorff perturbation

of the support—a much weaker requirement than convergence in any parametric metric—bounds the perturbation of the causal estimand. This delivers a robustness guarantee with no analogue in classical (parametric) causal inference. Kim and Lee [82] exploit precisely this transfer to establish stability of their TATE estimator under the silhouette-based summary $\mathcal{T} = \varphi(\cdot; r)$.

Proposition 3.2 (Identifiability from a topological functor)**.** *Let G vary over a class $\mathcal{G}$ of DAGs over the random vector X, let $\mathcal{P}_G$ denote the observational distribution induced by G, and let $\mathcal{T}$ be an admissible topological functor. Then G is identifiable from $\mathcal{T}(\mathcal{P}_G)$ alone—i.e., there is a function Ψ such that $\Psi(\mathcal{T}(\mathcal{P}_G)) = G$ for all $G \in \mathcal{G}$* if and only if *the assignment $G \mapsto \mathcal{T}(\mathcal{P}_G)$ is injective on $\mathcal{G}$.*

Sketch. The "only if" direction is immediate: if the map is non-injective, then two distinct $G_1 \neq G_2$ produce identical $\mathcal{T}(\cdot)$, so no inverse Ψ exists. The "if" direction is the definition of identifiability: an injective map admits a left inverse on its image, and any such left inverse plays the role of Ψ. Standard measurable selection (e.g., the Kuratowski–Ryll-Nardzewski theorem) supplies a measurable Ψ when $\mathcal{G}$ is countable, as in the finite-DAG case. □

Significance. Proposition 3.2 isolates the exact informational condition under which a topology-based discovery algorithm can hope to recover the underlying graph. It allows existing topology-based discovery procedures to be classified by what $\mathcal{T}$ they implicitly use: TOPIC [144] uses the score-Jacobian functor, SSTS [154] uses a mutual- information-based functor, and Hodge-decomposition methods [76] use the harmonic component of a 1-chain. The proposition does not, of course, certify that the injectivity condition *holds* for a given $\mathcal{T}$; we list this as Open Question 6.4 in Section 6.

Together, Propositions 3.1 and 3.2 establish that the TCDA tuple is non-vacuous: it inherits both the stability and the identifiability properties that motivate its definition. The remaining sections of this section show that the four domains of the taxonomy fall out naturally as the four "faces" of Definition 3.1—one for each of the components $\mathcal{T}$, $\mathcal{C}$ in their representational and robustness roles, plus one for the interpretive direction (reading causality *out* of $\mathcal{T}$) and one for the methodological direction (applying TCDA in specific data modalities).

3.2 Domain A: Topological Causal Representation

Domain A addresses the question: *which topological structure best represents the causal content of high-dimensional, non-Euclidean data?* The role of $\mathcal{T}$ in Definition 3.1 is here that of a sufficient statistic for the causal structure or for the interventional contrasts of interest. Three sub-domains arise.

A1. Persistent homology for causal structure learning.

Persistent homology of a multivariate distribution or its empirical dependence graph can serve as a sufficient statistic for the underlying causal DAG. Two threads converge here. First, modern *topological- ordering* approaches to causal discovery—e.g. TOPIC [144], SSTS [154], NoCurl-X [76], OptiTopo [153], and CaPS [123]—use rank-based or information-theoretic functors to extract a topological order from the data and then refine edges within that order; the order itself is a topological invariant of the joint distribution. Second, persistent homology of the *dependence network* of a multivariate time series identifies feedback cycles [48]; loops in PH_1 correspond to directed feedback loops in the underlying Granger graph, and a generative simulation framework [48] now exists to test how well TDA-based discovery methods recover known cycle structure. In the language of Proposition 3.2, sub-domain A1 chooses $\mathcal{T}$ to satisfy the injectivity condition for restricted classes of G (e.g. additive-noise models with non-Gaussian innovations).

A2. Mapper and Reeb graphs for causal abstraction.

Where A1 produces a graph that approximates G, A2 produces a *topological abstraction* of the underlying causal mechanism: a simplification of the data space along a filter function chosen to align with the causal structure. The Mapper algorithm [26, 128] partitions $\mathcal{M}$ along level sets of a filter f and glues the resulting clusters into a simplicial complex; under appropriate choices of f, the resulting complex captures coarse causal regimes (treatment-response strata, phase transitions, mechanism-shift points). Reeb graphs play the analogous role for continuous filters [16, 102]. The recent ball-mapper variant [43] is particularly attractive for causal abstraction because it bypasses the cover-selection problem and admits a stability guarantee. The statistical scaffolding of Lecci [89] applies directly: bootstrap confidence sets on cluster trees yield calibrated abstractions, which is essential when the abstraction is to serve as the carrier of a downstream causal query.

A3. Causal embeddings in persistence diagrams.

Sub-domain A3 represents the *causal effect itself* as a topological feature: each potential outcome distribution P^a is summarised by a persistence diagram $\mathrm{Dgm}_k(P^a)$ (or its silhouette $\varphi(\cdot; P^a, r)$) and the causal contrast is a distance, hypothesis test, or learned mapping between these diagrams. This is exactly the formulation of Kim and Lee [82], who define the *topological average treatment effect*

$$\mathrm{TATE} = W_1(\mathrm{Dgm}_k(P^{a=1}),\, \mathrm{Dgm}_k(P^{a=0})),$$

and provide an AIPW estimator with $\sqrt{n}$-rate. The same idea applies in non-Euclidean outcome spaces: the geodesic ATE of Kurisu et al. [88] for Riemannian-valued outcomes admits a topological generalisation in which the Fréchet mean is replaced by the persistence diagram. Sub-domain A3 thus exhibits the TCDA tuple in its most direct form: $\mathcal{T}$ converts outcomes to diagrams; $\mathcal{C}$ is a metric or test on those diagrams; the causal graph G underwrites identifiability of the contrast.

3.3 Domain B: Topological Robustness in Causal Inference

Domain B addresses the question: *when can topological invariants* ***control*** *the sensitivity of causal estimates to perturbations of the data, the model, or the identifying assumptions?* Here the role of $\mathcal{T}$ is to deliver Lipschitz-type bounds that propagate to $\mathcal{C}$.

B1. Stability of causal estimates under topological perturbations.

Proposition 3.1 is the prototype: any Lipschitz causal query whose primary statistic is a persistence diagram inherits a quantitative robustness bound from the bottleneck/Wasserstein stability of PH_k. Kim and Lee [82] exploit this in their stability theorem for the silhouette TATE; Farzam et al. [50] extend it to representation-learning settings where topological summaries serve as distributionally-robust losses. A complementary thread uses persistence landscapes [20] as smooth, Hilbert-space-valued embeddings, which allows classical M-estimation rates to be transferred to TDA-based causal estimators. The statistical inference programme of Lecci [89]—confidence sets, bootstrap, rates of convergence on landscapes and diagrams—is the inferential backbone of all B1 work: a TCDA estimator that lacks calibrated uncertainty cannot deliver causal robustness, only descriptive plots.

B2. Topological detection of unobserved confounding.

A persistent "hole" in the data manifold can betray the action of an unobserved variable: when an unobserved confounder U moves an outcome along an orbit, the support of the observed distribution exhibits a non-trivial PH_1 class that the observed graph cannot explain. This

$$
\begin{tikzcd}
P^{X=x} \arrow[r, "\mathcal{T}"] \arrow[d, "\mathrm{do}(x')/\mathrm{do}(x)"'] \arrow[d, dashed, bend left, "T_{\mathcal{C}}"] & \mathcal{T}(P^{X=x}) \arrow[d, "\Phi_{\mathcal{C}}"] \\
P^{X=x'} \arrow[r, "\mathcal{T}"'] & \mathcal{T}(P^{X=x'})
\end{tikzcd}
$$

Figure 2: Functorial picture of a TCDA query. The horizontal arrows apply a topological functor $\mathcal{T}$ (e.g. PH_k, Reeb, Map) to two interventional distributions. The left vertical arrow $\mathrm{do}(x) \to \mathrm{do}(x')$ acts at the level of the causal model; the right vertical arrow $\Phi_{\mathcal{C}}$ acts at the level of topological summaries. A TCDA solution requires either that the diagram *commutes* – so that the topological summary of a causal contrast equals the causal contrast of topological summaries – or that the failure to commute is itself a quantifiable, interpretable causal object. Stability of $\mathcal{T}$ (cf. Equation (4)) makes the right vertical arrow Lipschitz with respect to bottleneck distance; hence under commuting the left vertical arrow inherits a Lipschitz bound in the same metric.

observation has been operationalised in two ways. First, *anomaly scores* based on bottleneck or Wasserstein distance flag samples whose local persistence diagram differs from the population diagram [19, 107]; flagged samples are candidates for hidden-confounder action. Second, the topological hierarchy of Ibeling and Icard [72] shows that the set of SCMs for which an assumption-free identification of $P(y \mid \mathrm{do}(x))$ is possible is *topologically meager*, formalising the intuition that confounders are "everywhere dense." This puts a precise upper limit on what B2 detection methods can hope to achieve: they identify residual topology consistent with confounding but cannot, in general, certify the absence of confounding without further assumptions.

B3. Functorial causal inference.

Sub-domain B3 asks for the categorical analogue of Definition 3.1: a formal description of how causal inference and topological filtration interact as composable functors. The commutative square of Figure 2 captures the ideal case: applying $\mathcal{T}$ to a pair of interventional distributions and then taking a topological contrast should yield the same answer as taking the causal contrast at the level of distributions and then applying $\mathcal{T}$. Mahadevan [96, 97, 98] have developed a sheaf- theoretic framework in which causal inference is presented as a universal construction in a topos of sheaves over a base category of contexts; do-calculus is recovered through the subobject classifier. This programme can be read as a B3 formalisation of TCDA when $\mathcal{T}$ is chosen to be the global-section functor of an appropriate sheaf. The Alexandroff-topology approaches of Willig et al. [152] provide a more elementary entry point.

3.4 Domain C: Causal Interpretation of Topological Features

Domain C inverts the direction of arrows: rather than using topology to build or stabilise causal estimates, it reads *causal stories* off topological features that have already been computed. The role of $\mathcal{T}$ remains as before, but the role of $\mathcal{C}$ is now that of an interpretive overlay—a translation of a topological summary into the language of mechanisms, mediators, and heterogeneity.

C1. Betti numbers as causal explanations.

A persistent k-cycle is an invariant; whether it admits a causal interpretation depends on the data-generating process. In multivariate time-series neuroscience, a robust 1-cycle in the Granger-network filtration of a healthy brain corresponds to a re-entrant cortical loop that the disease modifies [48, 52, 129]. In gene-regulation, 1-cycles in correlation-network filtrations have been linked to feedback motifs whose perturbation alters phenotype [22, 86]. In each case, the topological feature is a candidate causal mechanism whose validity is adjudicated by intervention.

Sub-domain C1 develops the principled language for these moves: when is a persistent cycle "causal" rather than "spurious"?

C2. Persistent cohomology for mediation analysis.

Cohomology is more naturally suited to decomposition than is homology: a coboundary represents the action of a potential function on a chain, and the Hodge decomposition of a flow on a simplicial complex separates the gradient (direct), curl (feedback), and harmonic (global) components [41, 76]. We argue that this decomposition is the right topological tool for *mediation analysis*: the total causal effect of an exposure A on an outcome Y through mediators M decomposes naturally into a gradient component (direct effect), a curl component (cyclic feedback through M), and a harmonic component (effects not attributable to any local mediator). Open Question 6.6 in Section 6 states the corresponding open problem.

C3. Stratified spaces and causal heterogeneity.

Real causal effects are often heterogeneous across strata defined not by explicit covariates but by latent topological features: disease subtypes are connected components of a phenotype manifold; phase transitions are codimension-1 singularities. The theory of stratified spaces [8, 69] provides a framework for causal effects that vary across topological strata, generalising the conditional ATE to a stratum-conditional ATE indexed by a topological feature. The cluster-tree statistical inference of Lecci [89] and the topological clustering theory of Chazal et al. [32] provide the inferential machinery; the causal content remains to be developed and is one of the main open avenues we identify in Section 6.

3.5 Domain D: Methodology and Applications

Domain D is the applied face of TCDA. It does not introduce new mathematical primitives—these are supplied by Domains A–C—but it identifies the data modalities and substantive areas in which the TCDA tuple is most readily instantiated. Each application sub-domain specialises the tuple $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ to a class of problems.

D1. TCDA for time-series causality.

Time-series data are the natural habitat for sub-domain D1. The space $\mathcal{M}$ is a phase-space reconstruction (Takens embedding) of one or more observed series; G encodes temporal directed causality (Granger or its generalisations); $\mathcal{T} = \mathrm{PH}_1$ of the reconstructed phase-space detects periodicity and feedback structure; $\mathcal{C}$ is a hypothesis test or estimation of dynamical causality. Bando et al. [9] combine persistent homology of Takens embeddings with convergent cross mapping; Gholizadeh and Zadrozny [59], Harnack et al. [65] use persistence to augment Granger-type tests. El-Yaagoubi et al. [48] provide a simulation framework on which D1 methods can be benchmarked, with prescribed numbers of cycles serving as ground truth.

D2. TCDA for spatial and spatiotemporal causality.

The space $\mathcal{M}$ is a manifold ($\mathbb{R}^2$, $\mathbb{R}^3$, a sphere) or a spatio-temporal product; G encodes spatially-distributed causal mechanisms (e.g. teleconnections in climate, propagation in epidemics); $\mathcal{T}$ is typically cubical persistent homology [77] on a gridded scalar field. The cosmological literature [37, 134, 148] provides a mature template for descriptive TCDA on cosmic-web data; sub-domain D2 is the research programme that adds causal attribution to such descriptions (e.g. attribution of an extreme-precipitation event to a Pacific oscillation pattern via TCDA contrast). Kim and Lee [82]'s machinery extends naturally here when the outcome of interest is a gridded field rather than a vector.

D3. Industry applications: healthcare, finance, climate.

Sub-domain D3 collects the application-driven contributions that fall outside the cleaner D1/D2 modalities. Healthcare TCDA includes topological biomarkers of disease progression and treatment response [22, 74, 85, 86, 133], where Kim and Lee [82] exhibit the SARS-CoV-2 CT-scan application and the GEOM-Drugs molecular graph application. Finance TCDA includes topological early-warning indicators for systemic risk [61, 75], where the causal content is the attribution of a regime change to a triggering market factor. Climate TCDA includes topological characterisations of extreme-event clusters [37, 138], where the causal content is the attribution of regime change to forcing. In each case the TCDA tuple specialises predictably; the open methodological problem is the calibration of $\widehat{\mathcal{C}}_n$ under realistic non-stationarity.

3.6 Reading the framework

Figure 1 is intended to be read in two ways. *Vertically*, each domain A–D represents a different way to fix the TCDA tuple: A fixes $\mathcal{T}$ and $\mathcal{C}$ to representational roles; B fixes them to robustness roles; C inverts the direction by asking what $\mathcal{C}$ *means* given $\mathcal{T}$; D specialises $\mathcal{M}$ and G to particular data modalities. *Horizontally*, the sub-domains within each row form a methodological progression: from the simplest tool (homology/diagram) at left to the most structural (functorial/stratified) at right.

The framework is exhaustive in the following operational sense: every work we cite in this paper occupies at least one node of Figure 1, and most occupy more than one—reinforcing the claim that the field is real but currently fragmented. The framework is not, however, mutually exclusive; nor is it final. Section 4 populates the taxonomy with the actual state of the literature, and Section 6 returns to it to formulate the open questions and research avenues that each node generates.

We close this section with an observation that motivates the rest of the paper. Each of the four domains was identified by a single seed work: Ibeling and Icard [72] essentially establishes B2–B3 (the topology of SCMs and the meagre set of identifiable models); Kim and Lee [82] essentially establishes A3–B1 (causal contrasts as topological functionals with stability); the statistical programme of Lecci [89] provides the inferential machinery that A–C all require; and the time-series framework of El-Yaagoubi et al. [48] anchors D1 and supplies the simulation infrastructure on which the rest of the methodology can be benchmarked. The TCDA programme thus begins with four seed-points and the conjecture, defended in the remainder of this paper, that they are the corners of a single coherent research domain.

4 The State of TCDA: A Structured Literature Review

This section synthesises the current state of TCDA work, organised along the taxonomy of Section 3. We proceed chronologically through three broad eras—early connections (2015–2019), methodological advances (2020–2023), and application-driven work (2020–present)—and close with a structured account of remaining gaps. The four seed works of the field appear repeatedly as anchor citations: their role is to show that the methodological building blocks of TCDA are already in place and that the missing piece is the unifying framework we provide in Section 3. Figure 3 summarises the timeline.

4.1 Early connections (2015–2019)

The pre-2020 period saw TDA and causal inference develop largely independently. Three threads, however, foreshadow TCDA: the maturation of statistical TDA, the rise of topological approaches in causal-adjacent applications (brain networks, dynamical systems), and isolated calls for category-theoretic foundations on both sides.

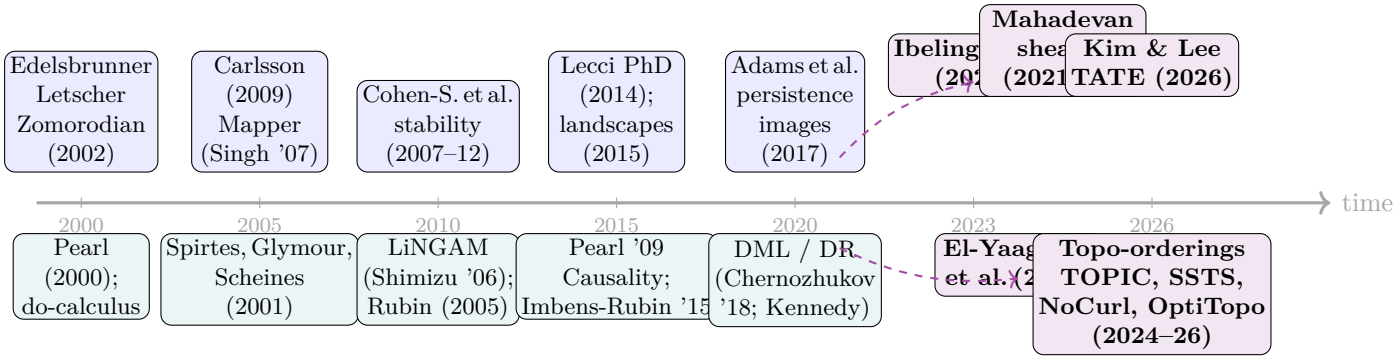

Figure 3: Parallel development of TDA (upper band, blue) and causal inference (lower band, teal) with emerging bridges (violet) in the 2020s. Until ∼2020 the two communities developed largely independently—TDA building from persistence and stability towards a statistical-inference programme [1, 23, 38, 47, 89], while causal inference matured under the Pearl and Rubin programmes and absorbed double-machine-learning [35, 73, 114]. The bridge papers of 2021–2026—topology on the space of SCMs [72], sheaf-theoretic causal calculi [98], topological treatment effects [82], simulation-based TDA inference for multivariate time series [48], and topological-ordering algorithms for causal discovery [76, 144, 153, 154]—together delineate the TCDA programme.

Statistical TDA matures. By 2014, the date of Lecci [89]'s PhD thesis-scale treatise, persistent homology had moved from algebraic topology and computational geometry into the statistical mainstream. Foundational algorithmic work [47, 156, 157] was complemented by a wave of inferential results: confidence sets for persistence diagrams via the bottleneck bootstrap [30, 51]; rates of convergence for the persistent homology of samples [28, 31]; Banach-space-valued landscapes that admit a functional central limit theorem [20, 33, 89]; and probability distributions on the space of diagrams [101, 145]. Lecci [89] is the most comprehensive PhD-thesis-scale synthesis of this material; the inferential machinery it organises—confidence sets, hypothesis tests, bootstrap, rates—is the statistical scaffolding that any subsequent TCDA estimator inherits. The introduction of persistence images [1], PerLay neural layers [27], and PLLay [84] closed the gap towards machine-learning workflows, and stability arguments for the silhouette [33] pre-figured the silhouette-based causal contrast of Kim and Lee [82].

Topological brain networks: causal-adjacent but not causal. Network neuroscience absorbed TDA early. The Bullmore-Sporns programme [12, 21, 137] cast brain connectivity as a graph, and TDA naturally lifted the discussion from graphs to higher-order simplicial structures [129, 130]. Cliques and cavities in the human connectome were shown to organise functional integration and segregation [129]; small-world propensity was re-cast as a topological invariant [103]; and trends in topological summaries of EEG/MEG across cognitive states began to serve as quantitative biomarkers [52, 62]. None of this work was framed in causal terms. Friston [53]'s functional and effective connectivity agenda was the closest causal counterpart, but it operated in a parametric (state-space, DCM) framework that did not interact with topology. The seeds of D1 (TCDA for time-series causality) and C1 (Betti numbers as causal explanation) of Section 3 are here visible in embryonic form.

TDA for time series and dynamical systems. The other natural site of TDA-causality interaction is dynamical systems: cycles in PH_1 of a Takens reconstruction correspond to recurrent or periodic behaviour. Chung et al. [36], Pachauri et al. [111] applied persistence to medical-image time series; Kim et al. [83], Umeda [147] used PH features for time-series classification; Gholizadeh and Zadrozny [59] surveyed the field. Stability of persistence under

filtration deformation [34, 38] ensured that the features used were not noise artefacts. Crucially, these works treated PH as a *descriptive* statistic; the next step—using PH inside a causality test—came after 2020 and is reviewed in Section 4.2.

Categorical and sheaf-theoretic preludes. On the structural side, Curry [40], Robinson and Turner [121], Robinson [122], Spivak [136] developed category- theoretic and sheaf-theoretic vocabularies for data analysis, emphasising functoriality and gluing. Independently, Pearl's [113, 114] programme had been re-cast in terms of categories of structural equation models, and Galles and Pearl [54] provided axiomatic foundations for counterfactuals. Henry et al. [68] explicitly argued that bridging the two would require new structural primitives. None of these papers connected TDA to causal inference; together they prepared the categorical vocabulary that the Mahadevan programme would later deploy.

Causal-inference foundations that TCDA will inherit. Concurrent with the TDA timeline, the causal-inference community consolidated around three programmes: Pearl's structural-causal-models and do-calculus [113, 114, 116]; Rubin's potential outcomes [73, 124, 125]; and the constraint-based and score-based discovery algorithms [100, 117, 127, 135]. A separate inferential thread on identification [4, 142, 143] established the language in which any TCDA solution must ultimately be phrased. The doubly-robust efficient-estimation programme [35, 79, 120] would, in turn, supply the AIPW machinery that Kim and Lee [82] extend to the topological setting in 2026.

The picture by 2019 is clear: both fields are mathematically and statistically mature; both have begun to be applied to the same empirical objects (brain networks, time series, cosmological structure); and yet no formal bridge yet exists. The four seed papers of the field all appear within the next four years.

4.2 Methodological advances (2020–2023)

Between 2020 and 2023 the bridge formed. Five sub-threads emerged in near-simultaneous fashion: (i) the topology of the space of structural causal models; (ii) category-theoretic and sheaf-theoretic causal calculi; (iii) topology-based causal discovery via ordering; (iv) causal treatment effects defined directly on topological summaries; and (v) topological deep learning with explicit causal applications.

The topology of SCMs. Ibeling and Icard [72] is the methodological pivot for Domain B of Section 3. The authors equip the space $\mathfrak{M}$ of structural causal models with a family of topologies that mirror the Pearl–Bareinboim causal hierarchy: the observational topology refines to an interventional topology and then to a counterfactual topology. The central theorem—a topological no-free-lunch result—states that the subset of $\mathfrak{M}$ on which assumption-free causal inference is possible is *topologically meagre* (in the sense of Baire). Two classical traditions converge here. The first is the measure-theoretic no-free-lunch programme of Belot [15], Oxtoby [110]; the second is the topology-of-statistical-verifiability programme of Dembo and Peres [42], Genin [55, 56], Genin and Kelly [57], Genin and Mayo-Wilson [58], which shows that open sets in the weak topology correspond to verifiable statistical hypotheses. Ibeling and Icard [72] lifts both to the causal hierarchy and provides accommodation for SCMs over countably-infinite variable sets via Borel-product topologies. The companion identifiability papers [70, 71] complete the picture. For TCDA, the result is foundational: it tells us that the search for a causal answer in the absence of topological structure on the model space is a search in a meagre set, and that topology is the right language for the negative results of the field.

Sheaf and topos foundations. A parallel programme, anchored in Mahadevan [96, 97, 98], recasts causal inference as a universal construction in a topos of sheaves over a base category of contexts. Do-calculus is recovered through the subobject classifier; identification becomes a sheaf-cohomological obstruction; the standard exchangeability/positivity/consistency triple is interpreted as descent conditions. This work realises B3 (functorial causal inference) of Section 3 and connects it to the broader categorical data-analysis programme of Curry [40], Hajij et al. [64], Papamarkou et al. [112], Spivak [136]. The Alexandroff-topology approach of Willig et al. [152] provides a more elementary entry point and recovers parts of the Ibeling-Icard hierarchy as Alexandroff specialisations.

Topology-based causal discovery: ordering algorithms. A different methodological thread takes the ordering of nodes in a causal DAG to be the primary object and infers it via topological or information-theoretic means. The score-based ordering of Rolland et al. [123] uses the Hessian of the log-density to extract a topological order under additive-noise assumptions and was followed by a wave of related work: TOPIC [144] uses topological-order constraints in a DAG-learning loss; SSTS [154] adapts the order to sequential and stochastic time-series; NoCurl-X [76] uses the Hodge decomposition of a 1-chain on the candidate DAG to extract the harmonic (acyclic) component; OptiTopo [153] optimises over topological orders directly; CaPS [123] originally introduced the score-Jacobian formulation. Hodge-decomposition methods more generally [41] ground the gradient/curl/harmonic split that we exploit in C2 (persistent cohomology for mediation analysis). In the language of Proposition 3.2, this entire thread selects a different topological functor $\mathcal{T}$ and asks when the $G \mapsto \mathcal{T}(\mathcal{P}_G)$ map is injective on a class of admissible graphs.

Treatment effects as topological functionals. Kim and Lee [82] is the methodological pivot for Domains A3 and B1. The authors define the *topological average treatment effect*

$$\mathrm{TATE} = W_1(\mathrm{Dgm}_k(P^{a=1}),\, \mathrm{Dgm}_k(P^{a=0}))$$

and a closely related power-weighted-silhouette contrast, develop a doubly-robust AIPW estimator with a $\sqrt{n}$-rate under product-rate nuisance conditions, prove weak-convergence guarantees in the space of silhouette functions, supply a W_1-based stability bound that links 1-Wasserstein perturbations of the diagram to sup-norm perturbations of the silhouette, and demonstrate the methodology on SARS-CoV-2 CT scans [133] and the GEOM-Drugs molecular-graph database [5]. The work is the first to combine all three ingredients of TCDA in one estimator: a non-Euclidean outcome space ($\mathcal{M}$ = a space of distributions over diagrams), a topological functor $\mathcal{T} = \varphi(\cdot; r)$, and a causal query $\mathcal{C}$ with calibrated uncertainty. Saki and Faghihi [126] extends the topological-contrast idea "beyond the silhouette" to more general functional summaries; Ballester et al. [7] develops complementary representations; Anonymous [2] connects the construction to deep causal models. The geodesic-ATE programme of Kurisu et al. [88] for Riemannian-valued outcomes is a sister contribution that places non-Euclidean potential outcomes on the same footing without invoking persistence.

Simulation-based inference and dependence networks. El-Yaagoubi et al. [48] addresses an under-served methodological problem: how to evaluate TDA-based inference for multivariate time series when no ground truth is available. The authors construct a generative model—mixtures of latent AR(2) processes—that allows researchers to *prescribe* the number of cycles in the dependence network, and they use this to test hypotheses about the resulting PH_1. El-Yaagoubi et al. [49] extends the construction to asymmetric dependence. Together these papers anchor D1 (time-series causality) of Section 3 and provide benchmarking infrastructure that the rest of the field can adopt.

Topological deep learning and topological regularisation. A fifth thread brings topology into representation learning. PerLay [27] and PLLay [84] are differentiable persistence layers that allow gradient-based learning on diagrams; the topological-deep-learning survey of Hajij et al. [64] consolidates the area, and Papamarkou et al. [112] argues for its centrality going forward. Farzam et al. [50] uses topological summaries as regularisers in causal-representation models, demonstrating robustness under heavy-tailed noise. Xu et al. [155] introduces topological invariants in deep generative models with causal structure. Guan [63] and Ma [94] extend these ideas to causal disentanglement.

The methodological state by 2023 is therefore that every component of Definition 3.1—a topological space $\mathcal{M}$ of observations, a causal graph G, a topological functor $\mathcal{T}$, and a causal query $\mathcal{C}$—is addressable by at least one well-developed line of research. What is missing is the joint study.

4.3 Application-driven work (2020–present)

The 2020–present period saw rapid uptake of TDA in scientific application areas where causal questions are central. We organise the account by domain.

Healthcare, drug discovery, and biomedicine. TDA of biological structure has a long pedigree [36, 78], but causal questions became central only recently. Persistent homology of protein conformations [22, 86] identifies binding-site topology that responds to ligand intervention—an instance of the TATE-style contrast in Kim and Lee [82], who explicitly use the GEOM-Drugs molecular-graph database [5] to demonstrate that ligand-binding induces topological change. CT-image analysis of SARS-CoV-2 [74, 133] constructs cubical persistent homology from imaging data and feeds the resulting summary into classification and treatment-effect estimators; Kim and Lee [82] use the same dataset as one of two methodological case studies. Topological feature extraction in tumour classification [74] and disease-progression modelling on cell-type manifolds [24, 39, 85] exemplifies the C3 (stratified heterogeneity) and D3 (industry applications) face of Section 3. A central remaining issue, which Section 6 treats as an open question, is identifiability when topological biomarkers are derived from imaging modalities subject to scanner and protocol drift.

Neuroscience and cognitive science. The brain-network literature [12, 21, 52, 103, 129, 130, 137] is the densest application site for sub-domain C1 (Betti numbers as causal explanation): a persistent 1-cycle in the functional Granger network of a healthy brain has been linked to re-entrant cortical loops that disease—epilepsy, Alzheimer's, schizophrenia—perturbs. Spectral and graphical extensions [62, 108] sharpen the picture. The simulation framework of El-Yaagoubi et al. [48] ties the empirical literature back to a controlled testbed: researchers can prescribe how many cycles a generative model produces and check that a TDA discovery method recovers them.

Climate, cosmology, and Earth science. Persistent homology of cosmological density fields [37, 134, 148] established the descriptive template for spatial TDA, and recent applications transfer this template to climate weather regimes [131, 132, 138]. The causal questions in this domain—attribution of an extreme-precipitation event to a Pacific oscillation, attribution of regime change to anthropogenic forcing—are precisely the spatiotemporal causal contrasts of sub-domain D2. To date, however, the TDA and the attribution communities have proceeded largely in parallel; the explicit TCDA pairing of $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ in climate science is nascent.

Finance and economics. Gidea and Katz [61], Ismail et al. [75] apply persistent-landscape analysis to financial time series and demonstrate that topological summaries serve as early-

warning signals of crashes. The causal content—attribution of a regime change to a triggering market factor—is implicit in their construction but not formally tested. Leykam and Angelakis [92] review the broader topological-finance literature.

Time-series and dynamical-system causality. Bando et al. [9] combine persistent homology of Takens embeddings with convergent cross mapping to extract dynamical causal links between coupled oscillators; Gholizadeh and Zadrozny [59], Harnack et al. [65] use persistence to augment Granger-type tests. Krasnovsky [87] extends this to multivariate stochastic systems with structural breaks. The combination directly instantiates the sub-domain D1 picture: $\mathcal{T} = \mathrm{PH}_1$ of a Takens embedding; G = Granger graph; $\mathcal{C}$ = a causal hypothesis test.

Materials and physical sciences. TDA in materials science [107] and topological photonics [92] pose causal questions about structure–property relationships. The TCDA tuple here selects $\mathcal{M}$ = a configuration space of atomic structures, $\mathcal{T} = \mathrm{PH}_{0,1,2}$ on the Vietoris-Rips filtration of atomic positions, and $\mathcal{C}$ a causal regression of property on structure mediated by topological features. This is one of the most natural application domains and is explicitly identified in Section 6.

4.4 Gaps in the current literature

Five gaps recur across the threads surveyed above. Each motivates a specific open question or research avenue in Section 6.

Absence of a unified framework. The five methodological threads of Section 4.2 fix different corners of the TCDA tuple but do not interact. Ibeling and Icard [72] study topology on $\mathfrak{M}$ but not topological summaries of data; Kim and Lee [82] study topological summaries of data but treat $\mathfrak{M}$ as fixed; Mahadevan [98] build the categorical scaffolding without explicit persistence-stability results; El-Yaagoubi et al. [48] provide benchmarks without a universal causal query. Our Definition 3.1 is the proposed unifier.

Computational and software fragmentation. TDA libraries (`Ripser` 13; `GUDHI` 141; `giotto-tda` 139) and causal-inference libraries (`DoWhy`, `causal-learn`, `EconML`) do not interoperate; converting a persistence diagram into a causal-effect estimand requires custom plumbing. Section 6 lists this as Computational Challenge C3.

Missing benchmark datasets. Causal-discovery benchmarks (CMU-CausalDisco, the CausalBench suite of 3, 150) do not include topological structure as a controlled covariate; conversely, TDA benchmarks do not encode causal mechanisms. The simulation framework of El-Yaagoubi et al. [48] is the closest analogue but is restricted to time-series dependence networks; a more general TCDA benchmark suite is one of the most actionable contributions a community project could make.

Theoretical disconnect between functorial and methodological threads. The Mahadevan sheaf programme [96–98] has not yet been used to analyse the AIPW estimator of Kim and Lee [82]; conversely, the Kim-Lee stability theorem has not been re-derived as a sheaf-cohomological obstruction. The two halves of the field speak different languages and the dictionary remains to be written. Several open questions in Section 6 (Q3, Q4, Q7) target precisely this disconnect.

Absence of formal causal interpretation for topological features. The brain-network and finance literatures freely use phrases like "a 1-cycle indicates a feedback loop" [48, 129] or "a persistent crash signature identifies a regime shift" [61, 75], but in neither case is the causal interpretation formally tied to an intervention. Sub-domain C1 of Section 3 exists precisely because the field has not yet articulated a principled language for assigning causal semantics to topological features. Section 6 formalises this as Open Question Q5.

The methodological building blocks of TCDA are therefore in place; what the field lacks is the joint formulation, the calibrated estimators, the interoperable software, the benchmark suites, and the explicit causal interpretive overlay on topological features. Sections 5 and 6 take these gaps as the starting point for the agenda we propose.

5 Case Studies and Proof-of-Concept Synergies

The case studies in this section are intended to demonstrate that the TCDA tuple of Definition 3.1 is not an empty abstraction: in each of five concrete settings we can write down the four components $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ explicitly, locate the example in the taxonomy of Figure 1, and indicate which parts are already established in the literature versus which constitute the gaps formalised in Section 6. The five cases were chosen to (i) cover all four top-level domains A–D of the taxonomy, (ii) engage each of the four seed works of the field—El-Yaagoubi et al. [48], Ibeling and Icard [72], Kim and Lee [82], Lecci [89]—in at least one case study, and (iii) span what we expect to be the most active sites of near-term TCDA research: biomedical imaging and molecular structure (Section 5.1), multivariate brain time series (Section 5.2), latent-variable confounding diagnostics (Section 5.3), Mapper-based causal abstraction with calibrated uncertainty (Section 5.4), and spatiotemporal climate attribution (Section 5.5). Table 1 summarises the mapping back to the taxonomy.

5.1 Case Study 1: Topological average treatment effects on biomedical imaging and molecular structure

Setting. Modern biomedical imaging and computational chemistry deliver non-Euclidean, high-dimensional outcome data—cubical CT volumes, volumetric MR scans, three-dimensional molecular point clouds—on which the classical ATE of eq. (6) is ill-posed because the outcome is not a scalar but a shape [5, 24, 74, 133]. The treatment effect of interest is a contrast between two shape distributions, not between two means. The first complete TCDA solution to this problem is the topological average treatment effect of Kim and Lee [82].

TCDA instantiation. We instantiate Definition 3.1 as follows. (i) $\mathcal{M}$ is the space of imaging or molecular outcomes, equipped with a natural ambient metric: the Hausdorff or Wasserstein distance on $\mathbb{R}^3$ point clouds for molecular data, the L^∞ pixel distance for cubical images. (ii) G is a small "treatment-to-outcome" DAG with a binary treatment $A \in \{0, 1\}$ (e.g. COVID-positive versus COVID-negative; ligand-bound versus apo conformation), pre-treatment covariates X (age, sex, scanner protocol, molecular descriptors), and the shape-valued outcome $Y \in \mathcal{M}$. (iii) $\mathcal{T}$ is composition of two stable functors: persistent homology PH_k—cubical for images [77], Vietoris–Rips for molecular point clouds—followed by the power- weighted silhouette $\varphi(\cdot; D, r)$ of eq. (5). By Lemma 2.1, $\mathcal{T}$ is Lipschitz from $(\mathcal{P}(\mathcal{M}), W_1)$ into $L^\infty(T)$. (iv) $\mathcal{C}$ is the *topological average treatment effect*

$$\mathrm{TATE} \;=\; \mathbb{E}[\varphi(\cdot;\, \mathcal{T}(Y^1), r)] \;-\; \mathbb{E}[\varphi(\cdot;\, \mathcal{T}(Y^0), r)], \tag{7}$$

together with the Wasserstein contrast $W_1(\mathrm{Dgm}_k(P^1), \mathrm{Dgm}_k(P^0))$ on the underlying diagrams.

Estimation and inference. Under the standard identifying assumptions (C1)–(C3) of Section 2.2, Kim and Lee [82] construct a doubly- robust augmented inverse-probability-weighting estimator for eq. (7) in which the outcome nuisance is a function-on-vector regression of $\varphi(\cdot; \mathcal{T}(Y), r)$ on (X, A) and the propensity nuisance is $\widehat{\pi}(x) = \widehat{\mathbb{P}}(A = 1 \mid X = x)$. Under product-rate nuisance conditions the estimator achieves a $\sqrt{n}$ functional rate in $L^\infty(T)$ and admits a tight weak-convergence limit, from which a level-α test of $H_0 : W_1(\mathrm{Dgm}_k(P^1), \mathrm{Dgm}_k(P^0)) = 0$ is read off. The statistical scaffolding inherits directly from Lecci [89]'s programme of confidence sets and bootstrap procedures for persistence diagrams, landscapes, and silhouettes; without this scaffolding, the TCDA estimator would deliver a point estimate but no uncertainty.

Stability. The estimator inherits the quantitative robustness of persistent homology via Proposition 3.1: a Hausdorff perturbation of the support of P^a translates into an L^∞ perturbation of the estimated silhouette TATE, with an explicit constant tracking the silhouette weight r. This is the concrete payoff of Domain B1: where classical AIPW estimators are sensitive to imaging artefacts and scanner drift, the topological AIPW estimator is provably robust to small Hausdorff perturbations of the support.

Why this case study matters and what remains open. The case study completely populates one "corner" of the TCDA taxonomy—sub-domains A3 and B1, with applications in D3—and is the clearest existence proof that the framework is non-vacuous: a non-Euclidean outcome that previously admitted only descriptive TDA now admits a calibrated, doubly-robust causal estimator. Remaining gaps targeted in Section 6 include the scaling of PH_k to volumetric three-dimensional inputs, transportability of the estimator across imaging protocols (a positivity issue), and the extension of the construction to non-binary or continuous treatments.

5.2 Case Study 2: Cycle-based causal discovery in multivariate dependence networks

Setting. Multivariate time-series data—neural recordings, financial returns, climate indices—admit a natural *dependence network* whose edges are weighted by pairwise statistical dependence (mutual information, partial coherence, transfer entropy). Causal questions on these networks concern the presence and direction of feedback loops: re-entrant cortical loops in EEG [48, 129], contagion cycles in financial markets [61, 75], oceanic–atmospheric teleconnections in climate [138]. Persistent homology of the dependence network in homology degree 1 is the natural detector of such loops.

TCDA instantiation. We set (i) $\mathcal{M}$ = the space of multivariate windowed time series in $\mathbb{R}^d$ with the supremum-distance topology; (ii) G = a directed Granger or transfer-entropy graph encoding lagged causal influence among the d variables; (iii) $\mathcal{T}$ = the composition "windowed dependence network" $\to$ "filtration by edge weight" $\to \mathrm{PH}_1$, optionally smoothed by the silhouette of eq. (5); (iv) $\mathcal{C}$ = a hypothesis test on the number of persistent 1-cycles in Dgm_1, or—more ambitiously—a recovery test for the cycle structure of G from Dgm_1. The construction is exactly the one analysed in El-Yaagoubi et al. [48], whose mixture of latent AR(2) processes allows researchers to *prescribe* the number of cycles in the network and so to evaluate $\mathcal{T}$-based discovery procedures against ground truth.

Inference and benchmark. El-Yaagoubi et al. [48] construct simulation-based confidence regions for PH_1 of the dependence network and use them to test hypotheses about cycle count. The companion work El-Yaagoubi et al. [49] extends the construction to asymmetric dependence and so to genuinely directed Granger graphs. The case study realises sub-domains A1 (PH for causal structure learning), C1 (Betti numbers as causal explanation), and D1 (time-series

causality) of Section 3 in a single, fully executable pipeline. Bando et al. [9], Gholizadeh and Zadrozny [59], Harnack et al. [65], Krasnovsky [87] pursue closely related strategies for dynamical causal links via PH of Takens embeddings, providing a complementary source of empirical evidence.

From a cycle to a causal mechanism. Sub-domain C1 raises the harder question: when does a persistent 1-cycle deserve the label "causal feedback loop" rather than the weaker label "cyclic correlation pattern"? The dependence-network construction does not, on its own, supply directionality. A natural answer, partial but concrete, is provided by Proposition 3.2 specialised to the dependence-network functor $\mathcal{T}$: if the assignment $G \mapsto \mathcal{T}(\mathcal{P}_G)$ is injective on a class of admissible graphs (e.g. acyclic Granger graphs with non-Gaussian innovations, in which case classical identifiability via the LiNGAM family applies [127]), then PH_1-based discovery is identifying. Where this injectivity condition fails—e.g. for two distinct cyclic graphs that yield the same windowed dependence network—the 1-cycle is a *joint* causal pattern of an equivalence class of graphs, and downstream identification requires auxiliary assumptions. The simulation framework of El-Yaagoubi et al. [48] makes both successful and unsuccessful cases controllable in the laboratory, and we list the characterisation of injectivity as an explicit open avenue in Section 6.

Why this case study matters and what remains open. The case study shows that an existing TDA benchmark is essentially a TCDA benchmark in waiting: equipping the dependence-network filtration with an explicit causal graph G converts the topological cycle detector into a topological causal-discovery procedure, with calibrated uncertainty inherited from Lecci [89]'s programme. The remaining methodological gap is the formal identifiability theory that would convert the construction from a benchmark-validated heuristic into a procedure with deductive guarantees on real (non-simulated) multivariate time series.

5.3 Case Study 3: Confounding diagnostics via persistent residual topology

Setting. Suppose an analyst posits a causal graph G over an observed set of variables and fits the corresponding SCM. Under correct specification the residuals are independent of the predicted endogenous variables and—crucially for TCDA—their joint support carries no topological information beyond Euclidean noise. When an unobserved confounder U acts on two or more observed variables along an unobserved orbit, the residual support inherits a persistent low-dimensional topology that no orientation of edges within G can explain. This observation is the basis of *topological confounding diagnostics* (sub-domain B2 of Section 3) and connects, via Ibeling and Icard [72], to the topological no-free-lunch theorem of the field.

TCDA instantiation. (i) $\mathcal{M}$ = the residual space of a fitted SCM, viewed as a point cloud in $\mathbb{R}^{|V|}$; (ii) G = the analyst's hypothesised graph, treated as a null hypothesis; (iii) $\mathcal{T} = \mathrm{PH}_1$ of the Vietoris–Rips filtration of the residual cloud, with bootstrap confidence sets [30, 51]; (iv) $\mathcal{C}$ = a test of the null hypothesis "the persistent 1-classes of $\mathcal{T}$ have lifetimes consistent with white-noise residuals."

Worked example: the circular confounder. Let U be uniformly distributed on the unit circle S^1 and define $X = \cos U + \epsilon_X$ and $Y = \sin U + \epsilon_Y$ with ϵ_X, ϵ_Y small, independent, mean-zero noise. An analyst who assumes no edge $X \to Y$ and no confounder will correctly estimate $\widehat{\mathrm{ATE}}(X \to Y) \approx 0$, since on average X carries no information about Y. The same analyst, however, will discover a persistent 1-cycle in Dgm_1 of the (X, Y) residuals whose lifetime exceeds any plausible noise band, signalling the action of U. Formally:

Proposition 5.1 (Topological confounding signature)**.** *Let U be a continuous random variable supported on a topological space $\mathcal{U}$ with non-trivial $H_1(\mathcal{U};\mathbb{R}) \neq 0$, and let $X_v = f_v(U, \epsilon_v)$ for $v \in V$ with f_v continuous and ϵ_v Gaussian and independent of U. Suppose the analyst's hypothesised graph G omits U and is otherwise correct. Then the limit support of the joint residuals admits a continuous surjection onto $\mathcal{U}$, and the persistent* PH_1 *of any sufficiently fine Vietoris–Rips filtration of the residuals contains at least one class whose expected lifetime is bounded away from zero uniformly in the noise scale.*

Sketch. By the continuous-mapping theorem and the assumed continuity of each f_v, the image of U under $v \mapsto X_v$ is a continuous surjection $\mathcal{U} \to \mathrm{supp}(P^X)$. The corresponding homology map $H_1(\mathcal{U}) \to H_1(\mathrm{supp}(P^X))$ is non-trivial whenever the (f_v) collectively are not homotopic to a constant, which is generic. The stability theorem (eq. (4)) ensures that the corresponding cycle persists in the empirical PH_1 of finite samples, with lifetime bounded below by the homological diameter of $\mathcal{U}$. Full details, including the rate at which the lifetime concentrates, follow the stability arguments of Chazal et al. [34], Cohen-Steiner et al. [38]. □

Discussion: detection is not identification. Proposition 5.1 is one-sided: a persistent residual 1-cycle is consistent with confounding but does not, on its own, recover U or even identify the variables it acts on. This asymmetry is exactly the content of the topological causal hierarchy theorem of Ibeling and Icard [72]: the set of SCMs admitting assumption-free identification of $P(y \mid \mathrm{do}(x))$ is topologically meagre, so almost any SCM carries residual topological structure that *flags* confounding but cannot *certify* its absence. The sheaf-theoretic programme of Mahadevan [96, 97, 98] provides a complementary reading in which the residual cycle is interpreted as a non-trivial class in a sheaf-cohomology obstruction to identification; this is the natural home of sub-domain B3.

Why this case study matters and what remains open. The case study converts an existing TDA descriptor—persistent 1-cycles of a point cloud—into a diagnostic that is logically linked to a causal property (residual confounding) via Proposition 5.1, and shows that the limits of the diagnostic are precisely those identified in Ibeling and Icard [72]'s meagre-set theorem. Operationalising the test on real data—calibrating the lifetime threshold against noise, controlling for nuisance topology induced by sampling, and extending to higher-dimensional confounders with $H_k(\mathcal{U}) \neq 0$ for $k \geq 2$—are the principal methodological gaps.

5.4 Case Study 4: Mapper-based causal abstraction with calibrated uncertainty

Setting. Many causal questions in medicine and the social sciences concern *regimes*: disease subtypes, treatment-responder strata, market phases. Regimes are typically not pre-specified by covariates but emerge from the geometry of the data. The Mapper algorithm [26, 43, 95, 128] produces a topological abstraction of a high-dimensional data set into a simplicial complex whose nodes are clusters of level sets of a chosen filter f; we propose that Mapper abstractions are the natural carrier of a regime-conditional causal estimand, and that Lecci [89]'s bootstrap-confidence programme delivers the calibrated uncertainty that regulated industries require.

TCDA instantiation. (i) $\mathcal{M}$ = a high-dimensional feature space of patient (or asset) descriptors. (ii) G = a coarse-grained DAG over Mapper nodes representing regimes and edges representing admissible transitions or treatment effects between regimes; we assume G is acyclic for identification, while permitting cycles inside the data Mapper graph itself. (iii) $\mathcal{T} = \mathrm{Map}_f$ with a filter f chosen to align with the causal mechanism (disease severity, time-to-event, an exposure score). (iv) $\mathcal{C}$ = a stratum-conditional ATE, defined for each Mapper node α by

$$\mathrm{CATE}(\alpha) \;=\; \mathbb{E}\big[\, Y^1 - Y^0 \mid X \in \alpha \,\big], \tag{8}$$

together with its uncertainty.

Bootstrap confidence sets for the abstraction. The statistical-TDA programme of Lecci [89] supplies the inferential scaffolding for $\mathcal{T}$ via the bottleneck bootstrap of Chazal et al. [30], Fasy et al. [51]: a confidence band $\mathcal{B}_n^{1-\alpha}$ around the empirical persistence diagram of the Mapper complex covers the population diagram with probability at least $1-\alpha$. Translating $\mathcal{B}_n^{1-\alpha}$ into uncertainty for eq. (8) proceeds in two stages. First, we bootstrap the Mapper graph to obtain a posterior over regime boundaries, propagating the cluster-membership uncertainty to each patient or asset. Second, we use the resulting node-level posterior weights as inputs to a regime-conditional AIPW estimator. The combination delivers, for each Mapper node α, a point estimate $\widehat{\mathrm{CATE}}(\alpha)$ and a confidence interval that accounts for both the topological (regime-assignment) and statistical (treatment-effect) sources of uncertainty.

Connection to stratified-space causal heterogeneity. The Mapper output is a stratified topological abstraction: nodes are 0-strata, edges are 1-strata, and higher simplices encode higher-order regime overlaps. The theory of stratified spaces [8, 32, 69] is exactly the right language for causal effects that vary across such strata—sub-domain C3 of Section 3. The bootstrap pipeline sketched above is therefore a concrete instantiation of Domains A2, B1, and C3 simultaneously.

Why this case study matters and what remains open. The case study addresses a critique frequently levelled at Mapper-based causal claims in healthcare and finance: the abstraction is descriptive, not inferential, and conveys no notion of statistical confidence. Lecci [89]'s programme dissolves the critique, provided the abstraction is treated as a TCDA functor. The open methodological question is the design of filter functions f that are simultaneously (i) causally informative—i.e. aligned with the mechanism of interest—and (ii) statistically well-behaved under the bootstrap. We list this as a research avenue in Section 6.

5.5 Case Study 5: Spatiotemporal causal attribution via cubical persistent homology

Setting. Climate-event attribution asks whether a specific extreme event would have occurred in a counterfactual climate without anthropogenic forcing. Conventional attribution operates on scalar summaries of the event (return period, intensity) and so is blind to the spatial structure of the extreme. Cubical persistent homology of the climate field captures that structure—the number of connected hot regions, the topology of pressure ridges, the persistence of low-pressure cells—and so naturally supports a topological notion of attribution [131, 132, 138].

TCDA instantiation. (i) $\mathcal{M}$ = the space of gridded scalar fields on $\mathbb{R}^2 \times [0, T]$ (daily precipitation, surface temperature, geopotential height), with the L^∞ pixel-wise distance. (ii) G = a coarse climate causal graph linking forcing (anthropogenic emissions, sea-surface temperature anomalies, modes of variability) to the spatial event field; the binary attribution treatment is "factual climate" versus "counterfactual climate" produced by a model ensemble. (iii) $\mathcal{T}$ = cubical PH_k for $k \in \{0, 1\}$ of the sublevel sets, with optional silhouette smoothing. (iv) $\mathcal{C}$ = a counterfactual contrast between persistence diagrams,

$$\mathcal{C}_{\mathrm{attr}} \;=\; W_1(\mathrm{Dgm}_k(P^{\mathrm{fact}}), \mathrm{Dgm}_k(P^{\mathrm{cf}})), \tag{9}$$

or its silhouette functional analogue analogous to eq. (7).

Table 1: Case studies and their place in the TCDA taxonomy of Figure 1. Each case study instantiates the TCDA tuple $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ of Definition 3.1 in a specific applied or theoretical setting, engages one or more of the four seed works of the field, and populates one or more sub-domains of the taxonomy.

§	Case study	Seed and bridge works	Taxonomy node(s)
5.1	Topological ATE on biomedical imaging	Axelrod and Gómez-Bombarelli [5], Kim and Lee [82], Lecci [89], Soares et al. [133]	A3, B1, D3
5.2	Cycle-based discovery in dependence networks	Bando et al. [9], El-Yaagoubi et al. [48, 49]	A1, C1, D1
5.3	Confounding diagnostics via residual PH	Fasy et al. [51], Ibeling and Icard [72], Mahadevan [98]	B2, B3
5.4	Mapper abstractions with calibrated uncertainty	Carrière et al. [26], Lecci [89], Singh et al. [128]	A2, B1, C3
5.5	Spatiotemporal climate attribution	Kim and Lee [82], Sousbie [134], Strommen et al. [138]	D2, B1, B2

The descriptive precedent and the causal upgrade. The cosmology literature [37, 134, 148] provides the mature descriptive template: persistence diagrams of cosmological density fields characterise the cosmic web, and bootstrap confidence sets on those diagrams supply calibrated descriptive uncertainty. The TCDA upgrade is straightforward in principle: replace cosmology's "observed universe vs. simulated universe" contrast by climate's "factual climate vs. counterfactual climate" contrast, and apply the AIPW machinery of Kim and Lee [82] treating each ensemble member as a sample from the relevant interventional distribution. The stability transfer of Proposition 3.1 controls the sensitivity of eq. (9) to grid resolution and to the small Hausdorff-scale perturbations that arise from regridding, bias correction, and interpolation.

Why this case study matters and what remains open. The case study realises sub-domains D2 (spatiotemporal causality) and B1 (stability of causal estimates) of Section 3 on a high-impact application. The principal remaining gap is identifiability of eq. (9) under unobserved natural variability: a coupled-model ensemble approximates the counterfactual distribution but does not eliminate the latent low-frequency oscillations that act as confounders. Proposition 5.1 suggests a built-in diagnostic: a persistent 1-cycle in the residual climate field *after* adjustment for the assumed forcing covariates would flag that some mode of variability has not been accounted for, and so the attribution estimate carries the residual topology of the unidentified causal structure. Climate TCDA is therefore both a showcase application and a stress test of the framework: its methodological success will depend on whether the diagnostics of sub-domain B2 can be deployed in concert with the estimators of sub-domains A3 and B1.

5.6 Synthesis: case studies and the taxonomy

Table 1 collects the case studies of this section, the seed and bridge works that anchor each, and the taxonomy nodes of Figure 1 that the case study populates.

Three observations emerge from the synthesis. First, every cell of the TCDA tuple $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ is exercised non-trivially in at least one case study: $\mathcal{M}$ ranges over image spaces, time-series spaces, residual spaces, feature spaces, and climate-field spaces; G ranges from simple treatment-to-outcome DAGs through dependence networks to coarse-grained regime graphs;

$\mathcal{T}$ ranges over PH_k in Vietoris–Rips, cubical, dependence-network, and Mapper forms; and $\mathcal{C}$ ranges over Wasserstein contrasts, silhouette ATEs, hypothesis tests, and stratum-conditional CATEs. The expressive scope of Definition 3.1 is therefore borne out empirically.

Second, the four seed works each anchor at least one case study, but no single seed work anchors more than two: the field is not the work of any one author or group, and a serious TCDA programme must draw on all four threads in parallel. The relative position of the seed works in the case-study landscape—Kim and Lee [82] as the estimator champion, Ibeling and Icard [72] as the no-free-lunch theorist, Lecci [89] as the statistical infrastructure, and El-Yaagoubi et al. [48] as the benchmark architect—is exactly the four-cornered division of labour anticipated in Section 3.

Third, every case study identifies a concrete remaining gap that we take up explicitly in Section 6: scaling, transportability, identifiability, filter design, and identification under unobserved variability. The case studies are therefore not endpoints but launch pads, and the agenda of Section 6 is calibrated against them.

6 Research Agenda and Open Questions

We close the technical part of the paper with a structured research agenda. The agenda is calibrated against the gaps identified in Section 4.4 and against the methodological openings left by the case studies of Section 5, and is organised along five branches: theoretical open questions (Section 6.1), methodological research avenues (Section 6.2), computational and algorithmic challenges (Section 6.3), domain-specific opportunities (Section 6.4), and industry- and implementation-facing problems (Section 6.5). Eleven formal Open Questions, fourteen Research Avenues, and four Computational Challenges are distributed across these branches; Figure 4 summarises the overall structure, and each item is formulated to be specific enough to underwrite a self-contained research project or grant proposal.

6.1 Theoretical open questions

The theoretical core of TCDA is the formal study of the four-tuple $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ of Definition 3.1 and of the conditions under which its components interact coherently. The questions below organise the agenda along the algebraic, categorical, statistical, and interpretive axes of that interaction.

Open Question 6.1 (Causal persistent modules)**.** Let $(P_t)_{t\geq 0}$ be a one-parameter family of interventional distributions induced by a continuously varying intervention $\mathrm{do}(x_t)$ on an SCM $\mathcal{M}$. Define a *causal persistent module* $M_\bullet^{\mathcal{C}}$ as the persistent homology of $(P_t)_{t\geq 0}$ with respect to a $\mathcal{C}$-graded filtration, so that birth and death scales encode the strength and duration of causal effects. Does $M_\bullet^{\mathcal{C}}$ admit a Crawley–Boevey-type decomposition into interval modules indexed by causal effects, and if so, do the resulting intervals coincide with classically identified causal estimands?

A positive answer would give an algebraic invariant of an SCM whose indecomposables are causal effects, in direct analogy with the way the indecomposables of an ordinary persistent module are barcode intervals. The question is open at every level: existence, uniqueness, stability, and computability.

Open Question 6.2 (Functorial adjoints for TCDA)**.** Let $\mathcal{T} : \mathbf{SCM} \to \mathbf{Pers}$ be a topological functor in the sense of Definition 3.1. Does $\mathcal{T}$ admit a left or right adjoint $\mathcal{T}^\sharp : \mathbf{Pers} \to \mathbf{SCM}$, and if so, what causal quantities does the unit $\eta : \mathrm{id} \Rightarrow \mathcal{T}^\sharp \circ \mathcal{T}$ or the counit $\epsilon : \mathcal{T} \circ \mathcal{T}^\sharp \Rightarrow \mathrm{id}$ compute?

Adjoints, if they exist, would systematically convert TDA-level data into the SCM that "best explains" it, providing a categorical foundation for inverse problems in causal discovery. The question sharpens the sheaf-theoretic programme of Mahadevan [96, 97, 98] and ties it to the functorial-data-analysis tradition of Curry [40], Hajij et al. [64], Spivak [136].

Open Question 6.3 (Causal stability theorem)**.** Let $\mathcal{M}_1, \mathcal{M}_2$ be SCMs in the weak Ibeling–Icard topology on $\mathfrak{M}$ and let $\mathcal{T} = \mathrm{PH}_k$. Suppose $d_{\mathrm{B}}(\mathcal{T}(p^{\mathcal{M}_1}), \mathcal{T}(p^{\mathcal{M}_2})) < \epsilon$. What is the tightest upper bound, as a function of ϵ and G, on

$$\sup_{x,y} \left\| P_{\mathcal{M}_1}(y \mid \mathrm{do}(x)) - P_{\mathcal{M}_2}(y \mid \mathrm{do}(x)) \right\|_{\mathrm{TV}} ?$$

This is the quantitative completion of Proposition 3.1: the proposition shows that any Lipschitz causal functional inherits robustness from bottleneck stability, but it leaves open the explicit bound for the central object of Pearl's framework—the interventional distribution itself. A positive answer would bridge the sheaf programme [98] and the AIPW programme [35, 82] by giving the "other half" of the dictionary: not just that diagrams perturb under data perturbation, but that do-effects perturb under diagram perturbation.

Open Question 6.4 (Identifiability of G from a topological functor)**.** For a fixed admissible topological functor $\mathcal{T}$ and a class $\mathcal{G}$ of DAGs, characterise necessary and sufficient conditions for the assignment $G \mapsto \mathcal{T}(\mathcal{P}_G)$ to be injective on $\mathcal{G}$. In particular: for which classes $\mathcal{G}$ is $\mathcal{T} = \mathrm{PH}_1$ of the windowed dependence network of El-Yaagoubi et al. [48] identifying?

Proposition 3.2 reduces TCDA identifiability to an injectivity question; Open Question 6.4 asks for the algebraic characterisation of that injectivity. The classical identifiability results for linear non-Gaussian models [127] and for score-based topological orderings [76, 123, 144, 153, 154] are the natural touchstones.

Open Question 6.5 (Causal semantics for persistent cycles)**.** Develop a calculus for assigning a causal label (*feedback*, *confounding*, *mediation*, *spurious*) to a persistent 1-cycle in $\mathrm{Dgm}_1(\mathcal{T}(X))$. Specifically, give sufficient observational and interventional tests that distinguish a cycle generated by a directed feedback loop in G from a cycle generated by an unobserved low-dimensional confounder.

This is the formal language whose absence is identified as a gap in Section 4.4. Proposition 5.1 supplies a one-sided test (residual cycle signals possible confounding); the question asks for the analogous tests for the other three cases and for the cohomological-charge calculus that would tie them together.

Open Question 6.6 (Persistent cohomology for mediation)**.** Let $A \to M \to Y$ be a mediation triple with a multidimensional mediator $M \in \mathbb{R}^d$. Apply the Hodge decomposition [41, 76] to a 1-chain on the simplicial complex generated by the empirical distribution of (A, M, Y) and decompose it into gradient (acyclic), curl (cyclic), and harmonic (globally non-trivial) components. Does the resulting decomposition recover the Pearl natural direct/natural indirect effect decomposition [114], and, if so, under what regularity conditions on the mediator distribution?

A positive answer would convert mediation analysis from a parametric exercise to a cohomological computation; the harmonic component, in particular, would quantify mediation that no local mediator captures.

Open Question 6.7 (Sheaf-cohomological reading of AIPW stability)**.** Re-derive the stability bound of Kim and Lee [82, Theorem on silhouette stability] as a sheaf-cohomological obstruction in the topos of Mahadevan [98]. Identify the obstruction class that vanishes if and only if the AIPW estimator on $\mathcal{T}$ inherits $\sqrt{n}$ rates under product-rate nuisance conditions.

Open Question 6.7 is the most concrete instance of the methodological dictionary called for in Section 4.4: the categorical and the asymptotic-statistical halves of TCDA must speak the same language before the field can be regarded as mature.

Open Question 6.8 (Functional CLT for TCDA beyond silhouettes)**.** Establish a functional central limit theorem for AIPW-type estimators of $\mathcal{C} = \mathbb{E}[\mathcal{T}(Y^1) - \mathcal{T}(Y^0)]$ for general admissible $\mathcal{T}$, including persistence landscapes [20], persistence images [1], and Euler characteristic curves. Identify the natural function space and the covariance structure of the limit.

The silhouette case is settled by Kim and Lee [82], but landscapes, images, and Euler curves carry different smoothness and sparsity properties and require their own central-limit machinery, parallel to but distinct from Bubenik [20]'s functional CLT for landscape means.

Open Question 6.9 (Positivity for non-Euclidean outcomes)**.** Reformulate the positivity assumption (C3) of Section 2.2 for outcomes valued in a non-Euclidean topological space $\mathcal{M}$. Is the classical pointwise overlap condition $0 < \mathbb{P}(A = 1 \mid X) < 1$ sufficient, or is a topological covering condition—e.g. each connected component of $\operatorname{supp} P^X$ intersects both treatment arms—also required for the topological contrasts of eqs. (7) and (9) to be identifiable?

Positivity violations occur silently when the outcome lives on a manifold or stratified space: a covariate stratum can have positive probability on one connected component of $\mathcal{M}$ and zero on another, without violating the classical scalar overlap. Open Question 6.9 asks for the topological refinement of overlap that detects this.

Open Question 6.10 (Transportability across topological domains)**.** Develop a transportability theory [10, 115] for TCDA estimators: under what conditions on $\mathcal{T}$, on the imaging or measurement modality, and on the causal graph G can $\widehat{\mathcal{C}}_n$ estimated on a source domain $(\mathcal{M}^{\mathrm{src}}, G)$ be transported to a target domain $(\mathcal{M}^{\mathrm{tgt}}, G)$ with bounded loss?

The case study of Section 5.1 motivates the question: the SARS-CoV-2 CT-scan TATE of Kim and Lee [82] is evaluated on a single scanner protocol, and the practical deployment of a TCDA estimator across heterogeneous clinical sites requires a transportability calculus.

Open Question 6.11 (Topological do-calculus)**.** Develop a topological do-calculus: a set of rewrite rules acting on persistence diagrams, landscapes, or silhouettes that mirror Pearl's three do-calculus rules and produce, in the limit, the same identification answers. Identify the topological analogues of the three Pearlian conditions (insertion/deletion of observations, exchanging actions and observations, and insertion/deletion of actions).

A topological do-calculus would extend the categorical work of Mahadevan [98] into a constructive algorithmic procedure, and would supply the inferential engine that Open Question 6.4 demands.

6.2 Methodological research avenues

The methodological agenda translates the theoretical questions of Section 6.1 into algorithms, estimators, and learning procedures. We enumerate five core avenues.

Research Avenue 6.1 (Causal Mapper)**.** Design a *causal* variant of the Mapper algorithm [26, 128] whose output is not a simplicial complex but a partially directed mixed graph (PDMG) over Mapper nodes, with edge orientations inferred from local conditional independence tests within each cluster. Provide bootstrap-based confidence guarantees on the orientation of each edge, building on Lecci [89].

The case study of Section 5.4 sketches the calibrated-CATE side of this avenue; Research Avenue 6.1 asks for the algorithmic upgrade that recovers a graph rather than a complex.

Research Avenue 6.2 (Persistent causal homotopy)**.** Develop a theory of *persistent causal homotopy* for time-varying causal systems: a two-parameter persistence framework [25, 91] in which one direction is the topological filtration scale and the other is causal time, and the bigraded module encodes both the topology of cross- sectional dependence and its temporal evolution.

El-Yaagoubi et al. [48] provide the empirical analogue (windowed dependence networks); Research Avenue 6.2 promotes the analogue to a calibrated bigraded estimator with formal interpretation in terms of regime changes and causal-structure shifts.

Research Avenue 6.3 (Topological regularisation for neural causal models)**.** Use topological summaries as regularisers in deep generative causal models, extending Farzam et al. [50], Guan [63], Ma [94], Xu et al. [155]. The objective is a causal variational autoencoder whose latent space is constrained to have prescribed persistent homology, encouraging causally meaningful disentanglement. Provide identifiability guarantees and downstream causal-effect estimation rates.

Research Avenue 6.4 (Multi-scale causal inference)**.** Treat the filtration parameter t of $\mathrm{PH}_k(\mathcal{F})$ as a *causal scale*: at fine scales the topology resolves micro-mechanisms; at coarse scales it resolves macro-mechanisms. Develop estimators of $\mathcal{C}$ as functions of t, with stability and identifiability guarantees that interpolate between the two regimes.

This avenue formalises an intuition long present in TDA—multi-scale structure is the field's defining property—in a causal language. The case study of Section 5.4 motivates the avenue: regime- conditional CATEs naturally depend on a scale.

Research Avenue 6.5 (Topological mediation via Hodge decomposition)**.** Implement the mediation decomposition of Open Question 6.6 as an estimator: compute the empirical Hodge decomposition of a 1-chain on the data simplicial complex, identify the gradient/curl/harmonic components, and map each to a causal estimand (direct, indirect, and non-local effect) with bootstrap confidence sets.

6.3 Computational and algorithmic challenges

Beyond statistical and mathematical theory, four computational problems gate the practical deployment of TCDA at the scale of modern scientific data.

Challenge 6.1 (Scalable persistent homology for causal-graph contexts)**.** Existing persistent-homology libraries [13, 139, 141] compute PH_k for $k \leq 2$ in $O(n^3)$ practical time for $n \lesssim 10^5$ points. Causal-graph applications routinely require $n \gtrsim 10^6$ (population-level omics, internet-scale finance). Develop sparse, approximate, or distributed PH variants whose accuracy guarantees suffice for the stability bound of Proposition 3.1 to remain informative at the resulting noise level.

Challenge 6.2 (Probabilistic persistent homology)**.** Calibrated uncertainty in TCDA requires probabilistic versions of PH_k [51, 99]. Develop Bayesian inference over persistence diagrams that scales beyond the small-n regime currently feasible and whose posterior contracts at the rate established by Lecci [89] for the bootstrap.

Challenge 6.3 (Interoperable software for TCDA)**.** The TDA stack (`Ripser`, `GUDHI`, `giotto-tda`) and the causal-inference stack (`DoWhy`, `causal-learn`, `EconML`, `CausalML`) are mutually opaque: there is no common data structure for a persistence diagram with attached causal- graph metadata, and no shared API for AIPW on a topological summary. Build a `tcda-py` (or analogous) library that exposes the TCDA tuple $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ as a first-class object and orchestrates estimation, inference, and reporting end-to-end.

Challenge 6.4 (GPU-accelerated cubical persistent homology)**.** Volumetric medical imaging (3D MR, 4D CT) and gridded climate fields require cubical PH_k on grids of size $\gtrsim 10^9$ voxels. Current cubical-PH implementations are CPU-bound and intractable at this scale. Develop GPU-accelerated, distributed cubical PH whose output is bit-for-bit compatible with the AIPW pipelines of Kim and Lee [82].

6.4 Domain-specific research opportunities

We collect five domain-specific avenues. Each is anchored in a case study of Section 5 and stated as a substantive, fundable programme of work.

Healthcare and drug discovery.

Research Avenue 6.6 (Topological causal models for drug repurposing)**.** Use persistent homology of disease-pathway networks and protein- ligand binding pockets [5, 22, 86] as the topological functor in a TCDA pipeline whose causal query is the predicted treatment effect of an off-label drug. The pipeline must produce a calibrated point estimate of efficacy together with a topological diagnostic for unmeasured comorbidities, generalising the Kim and Lee [82] GEOM-Drugs case study.

Climate and Earth-system science.

Research Avenue 6.7 (Topological attribution of extreme climate events)**.** Build a fully-specified TCDA pipeline that ingests an event-based climate ensemble (factual + counterfactual scenarios), computes cubical $\mathrm{PH}_{0,1}$ of the spatial event field, and outputs the counterfactual contrast of eq. (9) with bootstrap confidence sets. Demonstrate on a benchmark of canonical extreme events (Western European heatwave 2003, Pakistan flood 2022).

Finance and macroeconomics.

Research Avenue 6.8 (Topological causal early-warning systems)**.** Extend the topological landscape early-warning signals of Gidea and Katz [61], Ismail et al. [75], Leykam and Angelakis [92] into a causal early-warning framework: detect not only that a regime shift has occurred but identify the triggering causal factor via a topological contrast on the underlying network of asset returns, following Krasnovsky [87].

Neuroscience.

Research Avenue 6.9 (Whole-brain causal mapping via topological networks)**.** Synthesise the brain-network programme [21, 52, 129, 130] with the simulation-based inference of El-Yaagoubi et al. [48] to deliver whole-brain causal maps in which (i) cycles in the empirical PH_1 are formally tested against null Granger graphs and (ii) each surviving cycle is annotated with a putative causal mechanism by the calculus of Open Question 6.5.

Materials science.

Research Avenue 6.10 (Topological causal structure–property relations)**.** Apply the TCDA tuple to materials discovery: $\mathcal{M}$ is a configuration space of atomic positions, G encodes the causal mechanism linking structure to property, $\mathcal{T} = \mathrm{PH}_{0,1,2}$ of the Vietoris–Rips filtration of atomic positions [92, 107], and $\mathcal{C}$ is the predicted change in property under a topological intervention (e.g. introducing a defect). The output is a causal explanation of why a given microstructure produces a given macroscopic property.

6.5 Industry and practical implementation

The last four items in our agenda concern the translation of TCDA research into industrial practice. We state each as a Research Avenue because they require organised research programmes, not merely engineering.

Research Avenue 6.11 (Validation standards for topological causal models)**.** Develop reporting and validation standards for topological causal estimators in regulated industries (FDA-relevant clinical decision support, prudential regulators for financial early-warning systems). The standards must specify the minimum reportable uncertainty (the Lecci [89] bootstrap is a candidate baseline), the stability bound to be reported (a corollary of Proposition 3.1), and the confounding diagnostic threshold (a corollary of Proposition 5.1).

Research Avenue 6.12 (Stakeholder interpretability for topological causal findings)**.** Develop interpretability tools that translate topological causal findings—a persistence diagram, a Mapper graph, a counterfactual silhouette contrast—into the language of clinicians, policymakers, and managers. The constructive analogue of the Pearl "probabilities of causation" framework [6, 114] is a natural starting point: how does "a persistent 1-cycle implies a 75% probability of mechanism M" get reported?

Research Avenue 6.13 (Cloud-native infrastructure for large-scale TCDA)**.** Build a cloud-native compute layer that exposes the TCDA tuple $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ as a managed service, with built-in scaling for the GPU cubical PH of Challenge 6.4, distributed Vietoris–Rips of Challenge 6.1, and interoperable wrappers for the libraries named in Challenge 6.3. The deliverable is a reference implementation (open-source) and a managed-service blueprint (commercial).

Research Avenue 6.14 (Benchmark datasets for TCDA)**.** Design and release a TCDA benchmark suite analogous to the CMU-CausalDisco and CausalBench collections [3, 150] but specifically calibrated to topological-causal interactions: graphs with known homological signatures, treatment effects with known topological contrast, confounders with prescribed homotopy type. The simulation framework of El-Yaagoubi et al. [48] is the natural starting point; the suite must extend to imaging, molecular, and climate modalities.

6.6 Synthesis: a five-year and a ten-year horizon

Figure 4 aggregates the agenda. On a five-year horizon, the most actionable items are the *computational* challenges of Section 6.3—a working `tcda-py` library, GPU cubical PH, scalable Vietoris–Rips— together with the *methodological* avenues of Section 6.2 that are within reach of existing estimation theory: causal Mapper (Research Avenue 6.1), topological regularisation (Research Avenue 6.3), and Hodge mediation (Research Avenue 6.5). On a ten-year horizon, the field's foundational identity will be settled by the *theoretical* questions of Section 6.1: causal persistent modules (Open Question 6.1), functorial adjoints (Open Question 6.2), a quantitative causal stability theorem (Open Question 6.3), the identifiability characterisation (Open Question 6.4), and the sheaf-cohomological reading of stability (Open Question 6.7). The domain-specific opportunities of Section 6.4 and the industry items of Section 6.5 are the empirical loci where the field will earn its scientific standing.

Two observations close the section. First, every open question and research avenue connects back to a gap identified in Section 4.4 or to a remaining limitation noted in the case studies of Section 5: the agenda is not invented from first principles but is the systematic completion of the existing literature surveyed in Section 4. Second, the agenda is deliberately inclusive in scope and parsimonious in commitments: it declares which problems matter without prescribing which solutions will succeed, and so leaves room for a community of researchers from both TDA and causal inference to enter the field on their own terms. The next section—Section 7—returns to the overall vision and to the call for interdisciplinary collaboration that the agenda enables.

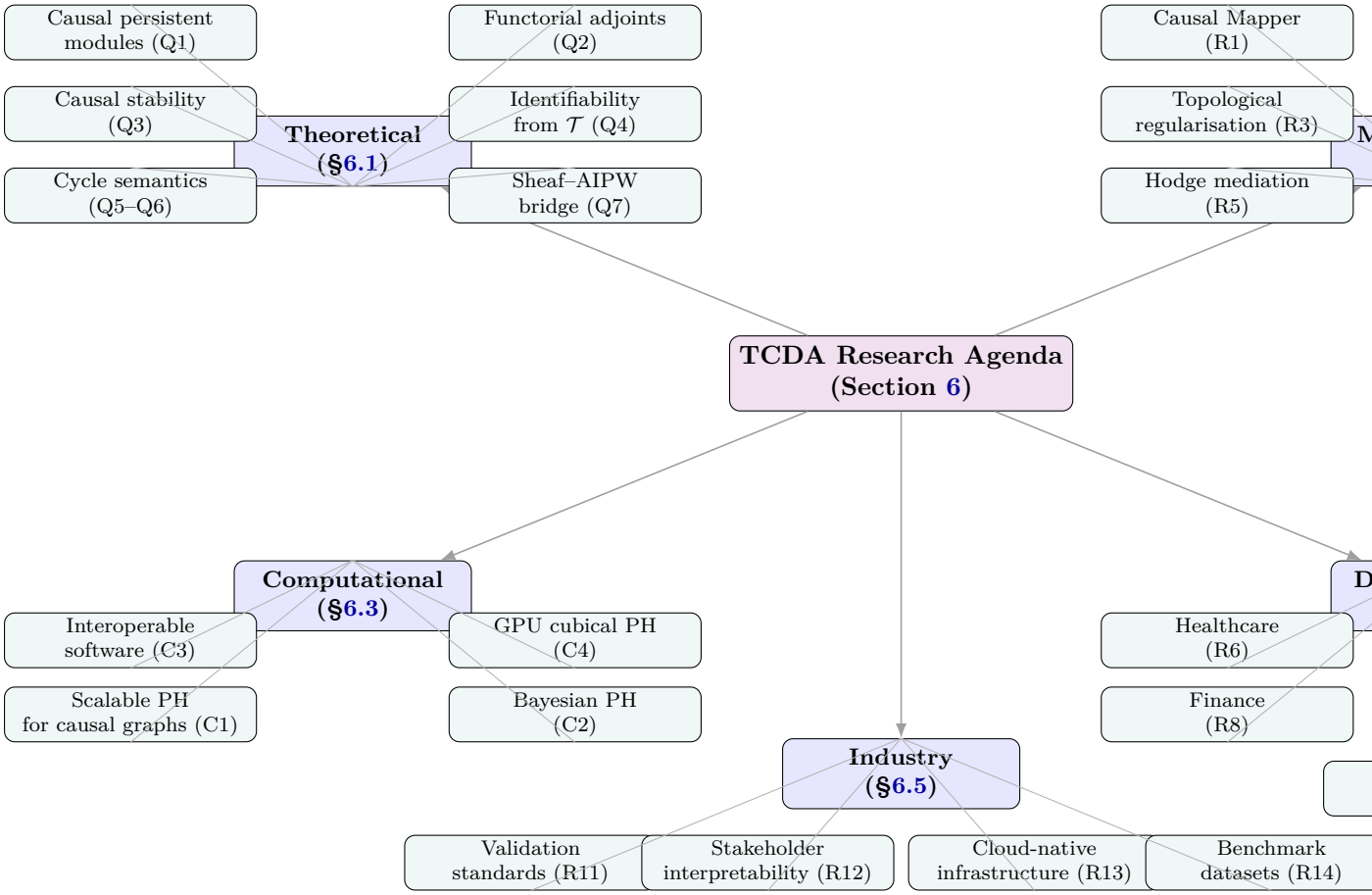

Figure 4: Mind-map of the TCDA research agenda. The agenda has five branches—theoretical (Section 6.1), methodological (Section 6.2), computational (Section 6.3), domain-specific (Section 6.4), and industry-facing (Section 6.5)—comprising eleven formal Open Questions (Q1–Q11), fourteen Research Avenues (R1–R14), and four Computational Challenges (C1–C4). Each leaf is keyed to the formal statement in the body of Section 6; together the leaves populate every domain of the TCDA taxonomy (Figure 1) with at least one concrete, grant-scale open problem.

7 Conclusion

We have proposed *Topological Causal Data Analysis* as a coherent research domain at the intersection of two mature mathematical traditions. The case for the proposal rests on four elements that the body of the paper has developed in turn: a formal framework (Section 3) anchored in the tuple $(\mathcal{M}, G, \mathcal{T}, \mathcal{C})$ of Definition 3.1 and the foundational propositions Propositions 3.1 and 3.2; a structured literature review (Section 4) showing that the methodological building blocks of TCDA are already in place but fragmented across five non-interacting threads (Section 4.4); a suite of five case studies (Section 5) demonstrating that the framework is non-vacuous and that each cell of its taxonomy admits an explicit instantiation; and a research agenda (Section 6) comprising eleven formal Open Questions, fourteen Research Avenues, and four Computational Challenges calibrated against the gaps and case-study limitations identified earlier.

The four cornerstones of the field. The argument of the paper is that the four seed works {Ibeling and Icard [72], Kim and Lee [82], Lecci [89], El-Yaagoubi et al. [48]}, taken together, span the mathematical, statistical, and methodological corners of TCDA. The first supplies the topology of causal-model spaces and the no-free-lunch theorem that frames the field's negative results [72]; the second supplies the doubly-robust, stability-guaranteed estimator

that converts a topological summary into a calibrated causal quantity [82]; the third supplies the statistical inference scaffolding—confidence sets, bootstrap, rates of convergence—without which the calibration of *any* TCDA estimator would be void [89]; and the fourth supplies the simulation-based benchmark on which discovery procedures in time-series and dependence-network settings can be evaluated [48]. No single seed work is the field on its own, and a serious TCDA programme must draw on all four threads in parallel.

What we have not done. We have not claimed that the framework is final, that every proposition is fully proved, that every open question is solvable on its current statement, or that the taxonomy is mutually exclusive. Several arguments in Sections 3 and 5 are proof sketches that demand fuller treatment, several research avenues in Section 6 are likely to fragment or merge as the field matures, and the case studies of Section 5 sit at varying levels of methodological readiness—some are recapitulations of published work, others are concrete proposals awaiting implementation. The four contributions enumerated in Section 1.4 should therefore be understood as a proposal in the technical sense—a unified vocabulary, a taxonomy, a literature map, and an agenda—rather than as a declaration of completion.

A call for interdisciplinary collaboration. The community most likely to settle the open questions of Section 6 does not yet exist. Its constituents—algebraic topologists, statisticians, machine-learning researchers, causal- inference theorists, computational scientists, and domain experts in biomedicine, climate, finance, neuroscience, and materials—are distributed across departments that historically have not shared seminars or journals. The contribution that TCDA can make beyond its internal technical agenda is to provide a vocabulary in which collaborations across these silos can be staged: a topologist can read Proposition 3.1 as a stability statement and contribute new admissible functors $\mathcal{T}$; a causal-inference theorist can read Proposition 3.2 as an identifiability question and contribute new classes $\mathcal{G}$; a domain scientist can read the case studies of Section 5 as templates and contribute new applications; a computational scientist can read Section 6.3 as a list of engineering targets. The challenges that result—the GPU cubical PH of Challenge 6.4, the interoperable software of Challenge 6.3, the benchmark suite of Research Avenue 6.14, the validation standards of Research Avenue 6.11—are themselves invitations to a kind of work that no single sub-community can complete in isolation. We hope this paper makes such invitations easier to extend and easier to accept.

Closing. The slogan we offer is simple: *the shape of data has a causal story to tell.* Topology, on its own, recovers the shape; causal inference, on its own, recovers the story; and only together can they deliver, for the increasingly complex and increasingly non-Euclidean data of contemporary science, the calibrated and intervention-aware accounts that practical deployment demands. The four contributions of this paper—a framework, a literature, a case-study suite, and an agenda—are offered as the starting point of that joint enterprise, not its conclusion. The body of work that converts the proposal into a mature field remains to be done, and it is to that body of work that we hope this paper now invites the reader.

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